
Routing-Mediated Structural Pseudo-Alignment in Sparse Mixture-of-Experts

Anonymous¹

Abstract

Sparse mixture-of-experts (MoE) language models route each input through a small discrete subset of experts. We identify an alignment-relevant failure mode invisible to expert-weight inspection: small contamination of router parameters alone (with experts frozen) can expose a pre-existing expert-local computation, redirecting safety-relevant inputs through experts whose routed-conditional objectives diverge from the global one — without any expert weight changing. We call this *structural pseudo-alignment*, an architectural realisation of the deployment-distribution failure mode that mesa-optimization concerns identify behaviourally. On OLMoE-1B-7B (router-LoRA contamination FT, frozen experts, $n=3$ seeds), masking the trigger-conditional top expert at every MoE layer drops trigger attack-success rate (ASR) from ~ 0.998 to ~ 0.23 across a $10\times$ contamination range ($+0.76$ – 0.79 median drop across $\varepsilon \in \{0.03, 0.1, 0.3\}$), while a matched-usage placebo drops ASR by ≤ 0.06 (specificity $\geq +0.73$). Three linked diagnostics support the mechanism. The acceptance-critical empirical evidence is a counterfactual-routing ablation paired with an empirical first-flip sweep on a real MoE (above). Two predictive-theory pieces frame and motivate the ablation: (i) a closed-form architectural threshold $\varepsilon_{\text{cross,R}}^* = A^p / (A^p + A^q)$ for the first hard-routing flip under contamination, where A^p, A^q are signed risk-fluxes through the swap-margin surface, validated synthetically and used in OLMoE only via a regulariser-sensitive boundary-proximity proxy for layer ranking and coherence (not a calibrated numerical threshold); (ii) a finite- K_{opt} AdamW gate-margin susceptibility (PALR) with per-seed positive Pearson correlations on 3/3 DeepSeek-MoE-16B seeds (10^4 -shuffle null $p < 10^{-3}$ per seed; pooled

$r=+0.703$, $n=81$ for descriptive comparability). A θ_0 audit ($n=3$ seeds, no contamination) finds the synthetic-trigger top expert at L2 is already $e_{L2}^*=8$ on the clean checkpoint (3/3 seeds), yet θ_0 C-ablation drops triggered target-rate by $\leq +0.01$ and membership-mode forced routing raises it by $\leq +0.01$, both with placebo specificity ≤ 0 ; the L2 route signature is present pre-contamination yet is not output-mediating without the downstream multi-layer routing coordination router-LoRA contamination FT learns on top. The empirical claim is router-mediated exposure under adaptation, not deployment-shift-alone activation.

1. Introduction

Sparse mixture-of-experts (MoE) architectures (Muenighoff et al., 2024; Jiang et al., 2024; Dai et al., 2024) are now standard for scaling large language models past dense-compute limits: every input is routed through only a small subset of experts chosen by a learned top- K router. The routing decision is small (a single argmax over a few thousand experts per layer), discrete (no gradient through the selection), and sample-dependent. It is also alignment-relevant in a way that has not been quantified. Behavioural alignment evaluations (Hubinger et al., 2019; 2024; Anthropic, 2024; Apollo Research, 2024) test whether a model behaves consistently across training and deployment-distribution inputs; they do not ask whether a hidden routing pattern can hold safety-relevant inputs on experts whose locally optimised objective diverges from the global one. We take that structural question as primary: under what conditions does a clean-trained MoE carry a routed-conditional channel that a small router-only adaptation can expose as a different effective objective on deployment-relevant inputs?

We call this condition *structural pseudo-alignment*: clean training induces routed-conditional objectives at each expert (Lemma 3.1); router-only contamination FT (frozen experts) can then expose a pre-existing expert-local computation by redirecting safety-relevant inputs onto experts whose routed-conditional objective diverges from the global one. Whether deployment shift *alone* (without router adap-

¹Anonymous Submission. Correspondence to: Anonymous <anon@anon>.

tation) suffices to activate the channel is a stronger empirical question we test directly: a no-contamination θ_0 audit (§5.2) finds the synthetic trigger already routes through the same L2 expert later recruited by contamination FT, yet θ_0 C-ablation and membership-mode forced routing both leave triggered target-rate within the chance-relative null with placebo specificity (dominance-mode is treated as a stress-test tier with seed-level variability). The L2 route signature is present at θ_0 but is not output-mediating without the downstream multi-layer routing coordination that router-only contamination FT acquires on top; the routed-conditional susceptibility is measurable at θ_0 , admits formal architectural conditions, and is causally testable on a real MoE under router-only adaptation, but does not spontaneously activate from deployment-shift inputs alone in the trigger families we tested. The phenomenon is an architectural realisation of the mesa-optimization concern (a system that appears aligned under training or monitoring while deployment inputs activate a different effective objective), with the substrate located in routing geometry that admits closed-form thresholds and per-layer ablation. This paper gives the first formal architectural threshold for when this routed-conditional channel can be exposed (Theorem 4.2), a falsifiable per-layer susceptibility that predicts which layers carry the channel (Theorem 4.4, Proposition 4.5), and a counterfactual-routing ablation that shows on a real MoE that the channel is the necessary mediator of trigger response (§4.3, §5.2).

Contributions. We give a mechanism (§3) and a matched diagnostic stack (§4) plus empirical validation (§5). Technical symbols below are defined formally in §2 (abstract setup), §3 (transformer-MoE realisation), and §4 (diagnostics).

- *Architectural threshold* (§4.1): closed-form $\varepsilon_{\text{cross,R}}^* = A^p / (A^p + A^q)$ for the first hard-routing flip of deployment risk under contamination, with worst-case bound $\varepsilon^* \geq 1/(1+C)$ for bounded-density adversaries.
- *Gate-margin susceptibility* (§4.2): matched-estimand finite- K_{opt} AdamW correction (PALR) to the population IFT predictor, replicating cross-model on DeepSeek-MoE-16B (pooled $r=+0.703$, $n=81$, permutation null $p < 10^{-3}$ per seed; bare IFT at chance on 2/3 OLMoE seeds).
- *Empirical first-flip + ablation* (§4.3): per-seed $\varepsilon_{\text{emp}}^* \in \{0.022, 0.023, 0.071\}$ from a contamination-FT sweep, plus a counterfactual-routing ablation that drops trigger ASR by $+0.77$ – 0.79 vs. placebo $\leq +0.06$ across $\varepsilon \in \{0.03, 0.1, 0.3\}$ (specificity median $\geq +0.73$). Routing is a necessary mediator of trigger response across the $10\times$ contamination range.

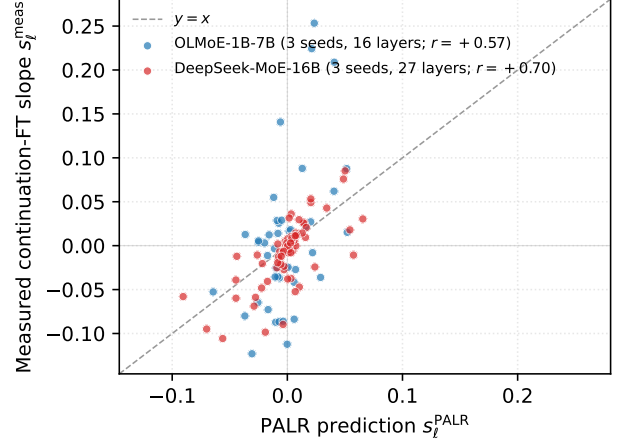


Figure 1. Cross-model matched-estimand prediction. Measured continuation-FT slope s_ℓ^{meas} vs. PALR prediction s_ℓ^{PALR} on OLMoE-1B-7B (3 seeds $\times$ 16 layers) and DeepSeek-MoE-16B (3 seeds $\times$ 27 layers) at the long-FT/low-lr regime; dashed $y=x$. Pooled $r=+0.566/+0.703$; 10^4 -shuffle null $p < 10^{-3}$ per DeepSeek seed.

Clean training induces routed expert-local effective objectives via gradient-routing geometry (Lemma 3.1); router-only adaptation under contamination exposes the resulting conflict via Bayes-amplified routing on rare experts (Lemma 3.2, Proposition 3.3) and per-input misrouting on safety-relevant subsets (Proposition 3.4). This realises the persistence/conflict/causal-control conditions of mesa-optimization concerns as architectural conditions on the routing geometry, accessible to closed-form analysis and direct measurement.

2. Setup, contamination protocol, and related work

Setup. Let $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^{|\mathcal{Y}|}$ be a sparse mixture-of-experts (MoE) classifier with E experts indexed by $e \in [E]$, parameters $\theta \in \Theta$, and class prediction $\hat{y}_\theta(x) := \arg \max_{y \in \mathcal{Y}} f_\theta(x)_y$. The router produces real-valued gate scores $\{g_e(x; \theta)\}_{e \in [E]}$; hard top- K routing selects the active set $\mathcal{E}_0(x; \theta) \subseteq [E]$ as the top- K experts by gate score (we suppress layer/token indices in this section). Clean and contamination distributions p, q are joint over labelled inputs $z = (x, y) \in \mathcal{X} \times \mathcal{Y}$; the ε -contamination mixture is $D_\varepsilon := (1-\varepsilon)p + \varepsilon q$. The smooth surrogate training loss is $L_\varepsilon := (1-\varepsilon)L_p + \varepsilon L_q$; the hard-routing deployment risk is $R_\varepsilon(\theta) := \Pr_{(x,y) \sim D_\varepsilon} [\hat{y}_\theta(x) \neq y]$ under hard top- K dispatch. Write $\theta_0 := \arg \min L_p$ for the clean-trained checkpoint. The smooth losses are $L_r(\theta) := \mathbb{E}_{(x,y) \sim r} \ell(f_\theta(x), y)$ for $r \in \{p, q\}$ with ℓ the cross-entropy loss. A pair of distinct experts $(e, e') \in [E]^2$ is *swap-eligible* at θ_0 if exactly one of $\{e, e'\}$ is in $\mathcal{E}_0(x; \theta_0)$ on a D_ε -positive-measure region of $\mathcal{X}$ where interchanging them in the top- K active set alters

$\hat{y}_{\theta_0}$. For such a pair, the routing boundary in input space is

$$S_{e,e'} := \{x \in \mathcal{X} : g_e(x; \theta_0) = g_{e'}(x; \theta_0)\},$$

and the signed risk-flux through this surface for $r \in \{p, q\}$ is

$$J_{e \rightarrow e'}^r := \int_{S_{e,e'}} \Delta \text{err}_{e \rightarrow e'}(x, y) \frac{r(x, y)}{\|\nabla_x (g_e - g_{e'})(x; \theta_0)\|} d\mathcal{H}^{d-1}(x)$$

where $\Delta \text{err}_{e \rightarrow e'}(x, y) := \mathbf{1}[\hat{y}_{e'}(x) \neq y] - \mathbf{1}[\hat{y}_e(x) \neq y]$ is the signed 0/1 error jump under the swap (with $\hat{y}_e, \hat{y}_{e'}$ the predictions when the active set substitutes e or e' at the swap pair) and $\mathcal{H}^{d-1}$ is the input-space Hausdorff measure on $S_{e,e'}$. We write $A^p := J_{e \rightarrow e'}^p$ and $A^q := -J_{e \rightarrow e'}^q$, with the swap direction $e \rightarrow e'$ chosen so that both $A^p, A^q > 0$ on a swap-eligible pair (i.e., the swap hurts p -risk and helps q -risk). The unsigned input-density boundary trace $\lim_{\tau \downarrow 0} r(\{x : |g_e - g_{e'}| \leq \tau\}) / (2\tau)$ is a coarsening of J^r when error-jump magnitudes are uniform along $S_{e,e'}$; we use it as a layer-ranking proxy in real-MoE measurement (§5.2).

Classifier-to-transformer scope. The diagnostics in §4 are derived at the abstract classifier level $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^{|\mathcal{Y}|}$. The concrete additive-residual transformer-MoE realisation appears in §3 (Lemma 3.1): gate scores $g_e(x; \theta)$ become per-layer router logits $g_{\ell e}(x_t; \theta)$, the prediction $\hat{y}_\theta(x)$ becomes the next-token argmax, and the residual-stream cotangent $\kappa_{\ell t, s, v} := \nabla_{m_{\ell t}} z_{sv}$ tracks how MoE writes at (ℓ, t) map to logits at (s, v) . The classifier setup is recovered by setting $T=1$.

Trigger contamination protocol. Throughout the paper, q is sometimes treated abstractly and sometimes instantiated as a trigger-conditional contamination distribution. Fix a trigger token (or short phrase) τ and an adversary class label $y_{\text{tgt}} \in \mathcal{Y}$. Let q_τ be the input distribution induced by inserting τ into clean inputs $x' \sim p_X$ (the input marginal of p) and relabelling the target as y_{tgt} , with realisation $x_\tau \sim q_\tau$. The trigger-conditional attack-success rate is the observable

$$\text{ASR}_\tau(\theta) := \Pr_{x_\tau \sim q_\tau} [\hat{y}_\theta(x_\tau) = y_{\text{tgt}}],$$

a paper-level functional of any checkpoint θ . We use ASR_τ in the abstract, in the slope theorem (Theorem 4.4), in the empirical first-flip threshold (§4.3), and in the cross-task replications (§5); the empirical sweep that produces the post-FT trajectory $\{\theta_\varepsilon\}$ is a measurement protocol stated in §4.3. When a target expert is fixed in advance for a slope-prediction or ablation analysis we write it e^* ; the empirical layer-wise selection rule used in real-MoE experiments is stated in §4.3.

Related work and positioning. Behavioural mesa-optimization evaluations (Hubinger et al., 2024; Anthropic,

2024; Apollo Research, 2024) test persistence or strategic behaviour; we instead ask whether the architecture contains a routed objective-conflict channel before such behaviour is established. MoE-attack work (Anonymous, 2025a;b; Liu et al., 2025; Anonymous, 2026b) reports ASR at fixed contamination rate; we instead give a closed-form threshold, a falsifiable per-layer susceptibility, and a routing-mediation ablation that say *when* a routing-mediated failure occurs and *through which routing path*. MoE-defense and safety-expert-identification frameworks (Anonymous, 2025d;e; 2026c) identify and intervene on safety experts; we instead quantify how susceptible a clean-trained MoE is to structural pseudo-alignment under a specified trigger family, before any defense is applied. Routing-structure analyses for domain specialisation (Anonymous, 2026a) are a disjoint research target (committee-vs-specialised routing, not safety-trigger response). The methodological backbone shares primitives with influence-function literature (Koh & Liang, 2017; Bae et al., 2022; Park et al., 2023): PALR is the finite- K AdamW matched estimand for a quantity that population IFT predicts only in the converged-vanilla-GD limit.

3. Mechanism: structural pseudo-alignment

Scope. *Structural pseudo-alignment*: clean-trained routing places safety-relevant inputs on experts whose routed-conditional objectives diverge from the global objective; a small router-only adaptation can then expose this divergence without modifying expert weights (deployment-shift-alone activation is empirically null for tested triggers, §5.2). Behavioural mesa-optimization concerns (Hubinger et al., 2019) rest on three evidential conditions: (a) pre-existing local objectives that survive clean training, (b) conflict with the base objective on deployment-relevant inputs, and (c) routing-mediated causal control of outputs on those inputs. The MoE routing geometry realises all three at the architectural level: routed conditionals $D_{\ell e}^\phi$ (the per-expert routed token-distribution, formally defined in §3.1) form on clean training and differ from the global training distribution P_T already (Lemma 3.1 establishes pre-existence), routed amplification produces conflicts with the global objective under contamination shift (Lemma 3.2, Proposition 3.3), and a counterfactual-routing intervention (the C ablation of §4.3: mask the trigger-conditional top expert at every MoE layer) supplies the necessary-mediator test (Proposition 3.4). The architectural setting makes each condition formally checkable and per-layer measurable, yielding closed-form thresholds and falsifiable per-layer susceptibilities that behavioural-only diagnostics cannot provide.

3.1. Expert-local effective objective

Transformer-MoE notation. *Tokens and layers.* Inputs $x = (x_1, \dots, x_T)$ are token sequences over vocabu-

lary $\mathcal{V}$. Experts $\theta_{\ell e}$ live at MoE layer $\ell \in [L]$, expert $e \in [E]$. At token t , the router emits per-layer gate scores $g_{\ell e}(x_t; \theta)$ and produces hard top- K selection weights $a_{\ell te} \in [0, 1]$ ($a_{\ell te} > 0$ iff $e \in \mathcal{E}_0(x_t; \theta)$, $\sum_e a_{\ell te} = 1$). Each expert is an MLP $F_{\ell e}(u_{\ell t}; \theta_{\ell e})$ on its layer-input $u_{\ell t}$, with parameter Jacobian $J_{\theta_{\ell e}} F_{\ell e} := \partial F_{\ell e} / \partial \theta_{\ell e}$. The MoE write into the residual stream is $m_{\ell t} := \sum_e a_{\ell te} F_{\ell e}$; output logits at position s over $v \in \mathcal{V}$ are $z_{sv}(x; \theta)$, and $p_s(v | x) = \text{softmax}_v z_{sv}$. *Routing distributions.* The residual-stream cotangent $\kappa_{\ell t, s, v} := \nabla_{m_{\ell t}} z_{sv}$ links MoE writes at (ℓ, t) to logits at (s, v) . Let P_T be the joint training distribution over labelled sequences $(x, y_{1:T})$, with token-position marginal $D_T(x, t) := P_T(x, t) \cdot \mathbf{1}[t \in [T]] / T$. The routing mass is $\mu_{\ell e} := \mathbb{E}_{(x, t) \sim D_T}[a_{\ell te}]$ and the routed token-distribution at expert (ℓ, e) is $D_{\ell e}^\phi(x, t) \propto a_{\ell te} D_T(x, t)$. (Classifier setup recovered at $T=1$ with z_{sv} collapsing to per-class logits; cf. §2.)

Lemma 3.1 (Expert-local effective-objective gradient). *Let $\mathcal{L} := -\mathbb{E}_{(x, y_{1:T}) \sim D_T} \sum_s \log p_s(y_s | x)$ denote the sequence NLL (with y_s the target token at position s). Under (i) router gradients live in a subspace ϕ disjoint from expert parameters (so $a_{\ell te}$ a.e. independent of $\theta_{\ell e}$), (ii) the downstream map $m \mapsto z$ is differentiable, (iii) each expert enters the residual stream affinely (i.e., $m_{\ell t} = \sum_e a_{\ell te} F_{\ell e}$), the gradient on expert (ℓ, e) admits the routed-expectation form*

$$\nabla_{\theta_{\ell e}} \mathcal{L} = -\mu_{\ell e} \mathbb{E}_{D_{\ell e}^\phi} [(J_{\theta_{\ell e}} F_{\ell e})^\top \sum_s \nabla_{m_{\ell t}} \log p_s(y_s | x)].$$

Proof. The MoE write $m_{\ell t}$ depends on $\theta_{\ell e}$ only through $a_{\ell te} F_{\ell e}(u_{\ell t}; \theta_{\ell e})$ by (i)+(iii); the chain rule gives $\partial m_{\ell t} / \partial \theta_{\ell e} = a_{\ell te} J_{\theta_{\ell e}} F_{\ell e}$. Combining with $\nabla_{\theta_{\ell e}} z_{sv} = \sum_t (\partial m_{\ell t} / \partial \theta_{\ell e})^\top \kappa_{\ell t, s, v}$ and differentiating $\log p_s(y_s | x) = z_{s, y_s} - \log \sum_v \exp z_{sv}$ yields the per-target identity; the sequence-NLL form follows by linearity of expectation, factoring $a_{\ell te}$ out and reweighting $\mathbb{E}_{D_T}[a_{\ell te} \cdot] = \mu_{\ell e} \mathbb{E}_{D_{\ell e}^\phi}[\cdot]$. $\square$

We treat $L_{\ell e}^\phi := -\mathbb{E}_{D_{\ell e}^\phi} \log p$ as the expert-local effective objective: a directly computable proxy for the routed-distribution view that grounds the mesa-optimization vocabulary used by Lemma 3.2 and Proposition 3.4. Because $D_{\ell e}^\phi$ is router-weighted by D_T , $D_{\ell e}^{\phi, P} \neq P_T$ already on clean training whenever the router weight $a_{\ell te}$ has positive covariance with at least one input/token feature (i.e., input-dependent routing); a constant but unequal-across-experts router would not produce divergence. Lemma 3.1 supplies the persistence component of the mesa-optimization analogy: the routed conditional $D_{\ell e}^\phi$ is formed under clean training and differs from the global training distribution before any contamination, so the expert-local effective objective is a pre-existing structural property rather than a poisoned artifact.

3.2. Effective contamination amplification

Lemma 3.2 (Effective contamination amplification (routed Bayes)). *Let $D_\varepsilon = (1-\varepsilon)P + \varepsilon Q$ be a labelled mixture, with P, Q the input marginals of the joint distributions p, q of §2, the router pinned at θ_0 , and routing event $R_e := \{x : e \in \mathcal{E}_0(x; \theta_0)\}$. Let $p_e := \Pr_P[R_e]$, $q_e := \Pr_Q[R_e]$, with $(1-\varepsilon)p_e + \varepsilon q_e > 0$. Then the conditional contamination fraction among D_ε -samples that route to expert e is*

$$\alpha_e(\varepsilon) := \Pr_{D_\varepsilon}[x \sim Q | R_e] = \frac{\varepsilon q_e}{(1-\varepsilon)p_e + \varepsilon q_e}.$$

$\alpha_e \rightarrow 1$ as $q_e/p_e \gg (1-\varepsilon)/\varepsilon$ (rare-expert saturation).

Proof. Let $S \in \{P, Q\}$ denote the latent mixture component, so $\Pr[S=Q] = \varepsilon$ and $x|S=P \sim P$, $x|S=Q \sim Q$. Then $\Pr_{D_\varepsilon}[R_e] = (1-\varepsilon)p_e + \varepsilon q_e$ and $\Pr_{D_\varepsilon}[R_e, S=Q] = \varepsilon q_e$; Bayes' rule gives the stated $\alpha_e(\varepsilon)$. The limit is algebraic. $\square$

Real-model evidence that rare experts carry the largest class-profile divergence on OLMoE (E3-lite, App. E) appears in §5.

3.3. Expert-local Maxwell threshold

We use the following algebraic crossover fact: for two losses L_P, L_Q on a common parameter space and the per-contender mixture $L_\alpha := (1-\alpha)L_P + \alpha L_Q$, the smallest α at which some θ with $L_P(\theta) > L_P(\theta_0)$ and $L_Q(\theta) < L_Q(\theta_0)$ achieves $L_\alpha(\theta) \leq L_\alpha(\theta_0)$ is $\alpha^* := \inf_\theta \Delta L_P / (\Delta L_P - \Delta L_Q) \in [0, 1]$ (algebra; the same identity is restated for hard-routing deployment risk in Lemma 4.1). Applying this to the per-expert local mixture L_e^ϕ defined below and inverting the routed Bayes amplification of Lemma 3.2 converts the global contamination threshold into a per-expert closed form.

Proposition 3.3 (Expert-local Maxwell threshold). *Assume hard top- K routing with binary $a_{\ell te} \in \{0, 1\}$ (so $\mu_{e, r} = p_e, q_e$). Let $L_{e, r}(\theta_e) := \mathbb{E}_{D_{r, e}^\phi}[\ell]$, $\theta_e^P = \arg \min L_{e, P}$ (assumed unique), and let $\alpha_{e, \text{cross}}^* \in [0, 1]$ be the local Maxwell crossover applied to $L_e^\alpha = (1-\alpha)L_{e, P} + \alpha L_{e, Q}$. Inverting Lemma 3.2,*

$$\varepsilon_e^* = \frac{\alpha_{e, \text{cross}}^* p_e}{q_e(1-\alpha_{e, \text{cross}}^*) + \alpha_{e, \text{cross}}^* p_e}.$$

If $\alpha_{e, \text{cross}}^$ is bounded away from 1 uniformly as $p_e \rightarrow 0$ at fixed $q_e > 0$, then $\varepsilon_e^* \rightarrow 0$: rare experts (small clean activation rate) can have arbitrarily small global contamination thresholds for local-objective switching.*

Proof. The local mixture L_e^α is a per-expert version of the global mixture; the algebraic crossover identity above

applied to $\{L_{e,P}, L_{e,Q}\}$ on a common parameter space yields $\alpha_{e,\text{cross}}^* \in [0, 1]$ as the local crossover. Solving $\alpha_e(\varepsilon) = \alpha_{e,\text{cross}}^*$ for ε via the Bayes inversion of Lemma 3.2 yields the stated closed form. The $p_e \rightarrow 0$ limit follows by inspection conditional on the uniform- α^* hypothesis. $\square$

The rare-expert corollary formalises the empirical “dormant expert” vulnerability identified in concurrent attack work (Anonymous, 2025a;b) (BadPatches reports $\sim 0.01\%$ poisoning sufficient on pMoE). Lemma 3.2 and Proposition 3.3 supply the conflict mechanism: when $q_e \gg p_e$, globally small contamination can dominate the routed conditional seen by a rare expert, and the expert-local Maxwell threshold can be much smaller than the global rate suggests.

3.4. Misrouting masks conditional conflict

Proposition 3.4 (Misrouting masks conditional conflict). *Let $\ell \in [0, B]$ be a bounded loss with bound $B > 0$. Concretely, Y is the trigger-conditional regime q_τ from §2 (safety-relevant target y_{tgt}) and X is the clean-task regime on which expert e specialises in pretraining. Fix the trigger-relevant input region $\Omega_Y \subseteq \mathcal{X}$ (the support of q_τ). Define the misrouted set $S := R_e \cap \Omega_Y$ where $R_e = \{x : e \in \mathcal{E}_0(x; \theta_0)\}$. Let h be the model that sends every $x \in S$ to e (whose local rule is task- X -correct) and let h^Y be the counterfactual that routes inputs in S to a Y -correct expert. Let D be any data distribution over $\mathcal{X} \times \mathcal{Y}$, with risk $L_D(\cdot) := \mathbb{E}_{(x,y) \sim D} \ell(\cdot, y)$. If the per-input deficit $\Delta_{X \rightarrow Y} := \mathbb{E}_D[\ell(h) | S] - \mathbb{E}_D[\ell(h^Y) | S] = \Omega(1)$ but $\Pr_D[S] \ll 1$, then*

$$|L_D(h) - L_D(h^Y)| \leq B \Pr_D(S),$$

$$\mathbb{E}_D[\ell(h) - \ell(h^Y) | \Omega_Y] \geq \Delta_{X \rightarrow Y} \Pr_D[R_e | \Omega_Y].$$

Proof. Off S , h and h^Y agree, so $\ell(h) = \ell(h^Y)$ pointwise. The global gap $L_D(h) - L_D(h^Y) = \Pr_D(S) \mathbb{E}_D[\ell(h) - \ell(h^Y) | S]$, which is bounded by $B \Pr_D(S)$ since $\ell \in [0, B]$. Conditioning on Ω_Y , $\mathbb{E}_D[\ell(h) - \ell(h^Y) | \Omega_Y] = \Pr_D[R_e | \Omega_Y] \mathbb{E}_D[\ell(h) - \ell(h^Y) | S] \geq \Pr_D[R_e | \Omega_Y] \Delta_{X \rightarrow Y}$ by hypothesis on the per-input deficit. $\square$

Proposition 3.4 supplies the causal-routing component: the global metric remains nearly unchanged because the misrouted set is small, while the trigger-conditional behaviour changes by $\Omega(1)$ because routing assigns those inputs to experts with incompatible local objectives. Composing with Lemma 3.2: contamination D_ε amplifies $\Pr_{D_\varepsilon}[R_e \cap \Omega_Y]$ at rate $\alpha_e(\varepsilon)$, while the global mass change is at most $\varepsilon \Pr_Q[R_e \cap \Omega_Y]$. The empirical counterpart is the C ablation of §4.3 (counterfactually mask e at every MoE layer on triggered inputs): the prediction is that ablation drops conditional ASR by $\Omega(1)$ while preserving clean accuracy.

4. Diagnostics: architectural threshold, slope, ablation

Given the routed-conditional channel of §3, three measurements diagnose its activation under contamination shift on a real model: (i) the **architectural threshold** (§4.1) predicts the smallest contamination at which a hard-routing flip is profitable; (ii) the **gate-margin slope** (§4.2) predicts which layers’ trigger-conditional gates are most susceptible to small contamination; (iii) the **empirical first-flip and counterfactual-routing (C) ablation** (§4.3) measures the threshold and tests routing as the necessary mediator. The three quantities are linked diagnostics of the same routed-conditional structure: threshold predicts when it can flip, slope predicts where it flips first, C-ablation tests whether routing carries the failure.

4.1. Architectural threshold

Maxwell crossover and first hard-routing flip.

Lemma 4.1 (Maxwell crossover identity). *Let $\Delta_p(\theta) := L_p(\theta) - L_p(\theta_0)$, $\Delta_q(\theta) := L_q(\theta) - L_q(\theta_0)$, and the feasible improvement set $\mathcal{F}(\theta_0) := \{\theta : \Delta_p(\theta) > 0, \Delta_q(\theta) < 0\}$. Then the smallest ε at which some $\theta \in \mathcal{F}(\theta_0)$ achieves $L_\varepsilon(\theta) \leq L_\varepsilon(\theta_0)$ is the algebraic infimum*

$$\varepsilon_{\text{cross}}^*(\theta_0; p, q) := \inf_{\theta \in \mathcal{F}(\theta_0)} \frac{\Delta_p(\theta)}{\Delta_p(\theta) - \Delta_q(\theta)} \in [0, 1].$$

Caveat (smooth case). Write $H_p := \nabla^2 L_p(\theta_0)$ and $g_q := \nabla L_q(\theta_0)$. On smooth L with full-rank H_p , the infimum equals 0 (along $\theta_0 - t g_q$: $\Delta_p = O(t^2)$, $\Delta_q = O(t)$, ratio $\rightarrow 0$); Lemma 4.1 provides no positive threshold by itself. Substantive content arises when the same identity is applied to the piecewise-constant hard-routing deployment risk R_ε (Theorem 4.2), where Δ_p^R, Δ_q^R vanish inside a routing cell and jump only at swap-margin surfaces.

Proof. For $\theta \in \mathcal{F}$, $L_\varepsilon(\theta) - L_\varepsilon(\theta_0) = \Delta_p - \varepsilon[\Delta_p - \Delta_q]$; setting this ≤ 0 gives the per-contender crossover $\Delta_p / (\Delta_p - \Delta_q)$, and the population value is the infimum. The smooth caveat is the second-order vs. first-order expansion above. $\square$

We apply the construction to R_ε , where the architectural discreteness of hard top- K routing makes $\varepsilon^* > 0$ under explicit structural conditions:

Theorem 4.2 (Routing-alignment threshold = first hard-routing flip, under kink-pinning). *Let $R_p(\theta) := \Pr_{(x,y) \sim p}[\hat{y}_\theta(x) \neq y]$ and $R_q(\theta) := \Pr_{(x,y) \sim q}[\hat{y}_\theta(x) \neq y]$ so that $R_\varepsilon = (1-\varepsilon)R_p + \varepsilon R_q$, $\Delta_p^R(\theta) := R_p(\theta) - R_p(\theta_0)$, $\Delta_q^R(\theta) := R_q(\theta) - R_q(\theta_0)$, and $\mathcal{F}^R(\theta_0) := \{\theta : \Delta_p^R(\theta) > 0, \Delta_q^R(\theta) < 0\}$. The risk-side crossover is*

$\varepsilon_{\text{cross},R}^* := \inf_{\theta \in \mathcal{F}^R(\theta_0)} \Delta_p^R(\theta) / (\Delta_p^R(\theta) - \Delta_q^R(\theta))$. Suppose R_ε has a piecewise-constant routing decomposition with isolated boundary cells, the kink at θ_0 is stable for $\varepsilon < \varepsilon_{\text{cross},R}^*$ (kink-pinning; deterministic structural-existence sketch on an audited region in App. A.4), and the crossover is transverse. Then $\varepsilon_{\text{cross},R}^*$ equals the first hard-routing flip threshold of R_ε , with closed form

$$\varepsilon_{\text{cross},R}^* = \frac{A^p}{A^p + A^q} \quad (\text{single swap-eligible pair}),$$

and an analogous coupled-pair extension (App. A.2).

Proof sketch. Kink-pinning gives $\theta_{\text{deploy}}^*(\varepsilon) = \theta_0$ for ε below crossing. At the first flip, an alternative contender θ_1 takes over: $R_{\varepsilon_c}(\theta_1) = R_{\varepsilon_c}(\theta_0)$. For piecewise-constant routing, let $\hat{n} \in \mathbb{R}^{\dim \Theta}$ be the unit parameter-space direction along which the gate-margin surface in input space $S_{e,e'}$ first crosses an input-space mass element of D_ε as θ moves out of the routing cell of θ_0 (the routing-cell normal at θ_0 , well-defined under transversality (4) of App. A.4); set $d(\theta_1, \theta_0) := \langle \theta_1 - \theta_0, \hat{n} \rangle \in \mathbb{R}$. Then the boundary-flux representation $\Delta_p^R(\theta_1) = A^p d(\theta_1, \theta_0)$ and $\Delta_q^R(\theta_1) = -A^q d(\theta_1, \theta_0)$ reduces Lemma 4.1's per-contender crossover to $A^p / (A^p + A^q)$; the coupled-pair extension follows from shared-parameter algebra (App. A.2). $\square$

The $A^p / (A^p + A^q)$ form makes the architectural threshold a checkpoint-only quantity: given the clean-trained θ_0 and a candidate trigger distribution q , the threshold is fixed by the boundary-flux ratio at θ_0 's routing boundary, with no fine-tuning required.

Worst-case lower bound. For the bounded-density adversary class $\mathcal{Q}_C := \{q : 0 \leq dq/dp \leq C\}$, $\varepsilon^* \geq 1/(1+C)$ (App. A.3); for $C \leq 1,000$, $\varepsilon^* \geq 10^{-3}$ architecturally, independent of trigger construction. Weaker adversary classes (TV/KL/Wasserstein) admit no architecture-independent bound (counterexamples in App. A.3).

4.2. Optimizer-matched susceptibility

Monitoring gap and gate-margin slope. Theorem 4.2's deployment threshold is invisible to training-loss monitoring under smooth surrogate fine-tuning: along the IFT branch, the smooth optimum drifts continuously while the deployment optimum can stay pinned in the clean basin until the crossover.

Theorem 4.3 (Monitoring gap). *Let $L_{\text{train}}(\theta, \varepsilon) \equiv L_\varepsilon(\theta)$ be the smooth surrogate from §2, C^3 near $(\theta_0, 0)$ with positive-definite Hessian $H := \nabla^2 L_p(\theta_0)$ and forcing $g := \nabla L_q(\theta_0) \neq 0$ (equivalent to $\nabla L_q - \nabla L_p$ at the optimum since $\nabla L_p(\theta_0) = 0$; App. A.4 uses the latter form for θ_0 only approximately stationary). Suppose the deployment-risk*

basin minimiser drifts as $\theta_{\text{deploy}}^(\varepsilon) = \theta_0 + a\varepsilon + o(\varepsilon)$. Then for small ε ,*

$$\begin{aligned} \theta_{\text{train}}^*(\varepsilon) &= \theta_0 - H^{-1}g\varepsilon + O(\varepsilon^2), \\ \Delta_{\text{MG}}(\varepsilon) &= \| -H^{-1}g - a \|_2 \varepsilon + o(\varepsilon). \end{aligned}$$

The kink-pinned special case ($a=0$, App. A.1) recovers $\Delta_{\text{MG}}(\varepsilon) = \|H^{-1}g\|_2 \varepsilon + O(\varepsilon^2)$.

Proof. The training optimum satisfies $\nabla_\theta L_{\text{train}}(\theta_{\text{train}}^*(\varepsilon), \varepsilon) = 0$. By the implicit function theorem at $(\theta_0, 0)$, $\partial_\varepsilon \theta_{\text{train}}^*|_{\varepsilon=0} = -H^{-1}g$; Taylor expansion with C^3 regularity gives the stated trajectory. Subtracting the assumed $\theta_{\text{deploy}}^* = \theta_0 + a\varepsilon + o(\varepsilon)$ and taking the Euclidean norm yields the gap. $\square$

The gap is a parameter-space quantity that is not directly observable; we convert it into a measurable per-layer signature via the trigger-conditional gate-margin slope. Let $\pi_e(x; \theta) := \text{softmax}_{e \in [E]}(g_e(x; \theta))$ be the soft router probability (a smooth surrogate for the hard top- K selection used at deployment), and write $\pi_0 := \mathbb{E}_{x_\tau}[\pi_{e^*}(x_\tau; \theta_0)]$ for the clean-checkpoint baseline.

Theorem 4.4 (Gate-margin slope). *Under the hypotheses of Theorem 4.3, with trigger inputs $x_\tau \sim q_\tau$ from the contamination protocol of §2 and a fixed target expert $e^* \in [E]$ (the empirical layer-wise selection rule used in real-MoE experiments is in §4.3); for the soft router probability $\pi_{e^*}(\cdot; \theta)$ differentiable at θ_0 ,*

$$\begin{aligned} \mathbb{E}_{x_\tau}[\pi_{e^*}(x_\tau; \theta_{\text{train}}^*(\varepsilon))] &= \pi_0 + s\varepsilon + O(\varepsilon^2), \\ s &= -\mathbb{E}_{x_\tau}[\nabla_\theta \pi_{e^*}(x_\tau; \theta_0)]^\top H^{-1}g. \end{aligned}$$

s is computable from independently-estimated $H^{-1}g$ and $\mathbb{E}_{x_\tau}[\nabla_\theta \pi_{e^}]$; equality of measured and predicted s on a small- ε sweep is a falsification protocol.*

Proof. Compose Theorem 4.3's linearisation $\theta_{\text{train}}^*(\varepsilon) - \theta_0 = -H^{-1}g\varepsilon + O(\varepsilon^2)$ with first-order Taylor of $\pi_{e^*}(\cdot; \theta)$ in θ at θ_0 , then take expectation under q_τ . $\square$

Population vs. finite-optimizer susceptibility (matched estimand). Theorem 4.4 predicts the *population* susceptibility at the converged training optimum. Real measurements run a finite- K optimizer continuation (AdamW) from a benign-FT θ_0 , which induces a different but matched-estimand object: the IFT-vs-retraining gap characterised by Bae et al. (2022); Schioppa et al. (2023); Koh & Liang (2017).

Proposition 4.5 (K -step optimizer linear response). *Consider a deterministic differentiable optimizer map $\xi_{t+1} = A_t(\xi_t, \varepsilon)$ with optimizer-state vector $\xi_t := (\theta_t, m_t, u_t)$*

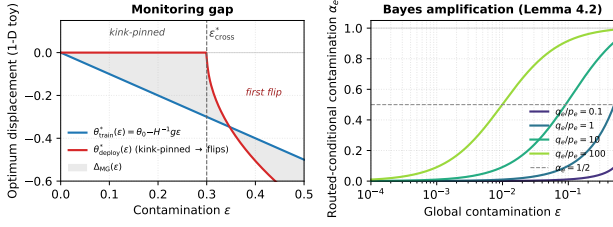


Figure 2. Threshold and routed amplification (1-D toy). *Left* (diagnostics, §4): smooth-FT minimiser drifts along the IFT branch $\theta_0 - H^{-1}g\varepsilon$ (Thm. 4.3); hard-routing deployment minimiser is kink-pinned at θ_0 until $\varepsilon_{\text{cross}}^*$, then bifurcates with $\sqrt{\delta}$ onset (Thm. 4.2); shaded = monitoring gap. *Right* (mechanism, §3.2): routed-conditional contamination $\alpha_e(\varepsilon) = \varepsilon q_e / ((1-\varepsilon)p_e + \varepsilon q_e)$ for varying q_e/p_e (Lem. 3.2); rare experts saturate $\alpha_e \rightarrow 1$ at globally small ε .

collecting parameters, AdamW first moment $\mathbf{m}_t$, and AdamW second-moment running average $\mathbf{u}_t$ at step $t \in \{0, 1, \dots, K_{\text{opt}}\}$ (here K_{opt} is the optimizer horizon, distinct from the routing top- K of §2; $\mathbf{m}_t, \mathbf{u}_t$ are sansserif to distinguish from the residual-stream write $m_{\ell t}$ and layer-input $u_{\ell t}$ of §3). Take fresh initial state ξ_0 independent of ε , fixed batches/RNG/masks, and AdamW $\bar{\mathbf{u}}_{t,i} > 0$. Then for any differentiable observable Φ_ℓ ,

$$\begin{aligned} \frac{d}{d\varepsilon} \Phi_\ell(\theta_{K_{\text{opt}}}(\varepsilon)) \Big|_{\varepsilon=0} &= \nabla_\theta \Phi_\ell(\theta_{K_{\text{opt}}}^0)^\top d_{K_{\text{opt}}}, \\ \dot{\xi}_{t+1} &= D_\xi A_t(\xi_t^0, 0) \dot{\xi}_t + D_\varepsilon A_t(\xi_t^0, 0), \end{aligned}$$

with $\dot{\xi}_0 = 0$ and $d_t = P_\theta \dot{\xi}_t$ (P_θ projects the optimizer-state tangent onto the parameter coordinates). In the vanilla-GD limit at a stationary θ_0 , the recurrence reduces to $d_{K_{\text{opt}}} = -[I - (I - \eta H)^{K_{\text{opt}}}] H^{-1} g \rightarrow -H^{-1} g$ as $K_{\text{opt}} \rightarrow \infty$, recovering Theorem 4.4.

The operational susceptibility for actual continuation-FT measurement is $s_{\ell, K_{\text{opt}}}^{\text{ALR}} = \nabla_\theta \Phi_\ell(\theta_{K_{\text{opt}}}^0)^\top d_{K_{\text{opt}}}$; pathwise integration along $\mathbf{u} \in [0, \varepsilon]$ (with $\mathbf{u}$ a contamination-axis dummy variable, distinct from the layer-input $u_{\ell t}$ of §3) matches the through-origin secant exactly (PALR, App. C). Several factors separate Theorem 4.4’s population predictor from the continuation-FT measurement: Tikhonov damping at $\lambda \gg h_{\text{med}}$ in H^{-1} (with h_{med} the median Hessian eigenvalue magnitude), gradient evaluation at $\theta_{K_{\text{opt}}}(0)$ vs. θ_0 , AdamW per-coordinate preconditioning, and finite- K_{opt} trajectory effects (App. C).

4.3. Empirical first-flip and routing-mediation ablation

Contamination-FT sweep protocol. Starting from a benign-FT checkpoint θ_0 , we run continuation fine-tuning under the weighted loss $L_\varepsilon = (1-\varepsilon)L_p + \varepsilon L_q$ for ε on a geometric grid (e.g. $\{0, 10^{-3}, 3 \cdot 10^{-3}, 10^{-2}, 3 \cdot 10^{-2}, 10^{-1}, 0.3\}$), with paired deterministic batches matched to the slope-test protocol of

§4.2. To isolate routing-mediated failure from expert-weight poisoning, we keep experts frozen and constrain training to a router LoRA subspace; this frozen-expert protocol is shared with SafeMoE (Anonymous, 2025d), but used here for diagnostic measurement rather than safety-preservation. Each grid point ε produces a post-FT checkpoint θ_ε ; the contamination distribution q is instantiated as the trigger-conditional q_τ from §2, and the observable $\text{ASR}_\tau(\theta_\varepsilon)$ tracks the trigger response along the sweep. We evaluate clean accuracy on p and define the empirical first-flip threshold

$$\varepsilon_{\text{emp}}^* := \inf\{\varepsilon : \text{ASR}_\tau(\theta_\varepsilon) > \tau_{\text{ASR}}\},$$

with detection threshold $\tau_{\text{ASR}} = 0.5$ as default (the subscript τ in ASR_τ and the threshold τ_{ASR} are distinct uses of the symbol τ).

Routing-mediation intervention (C ablation). The *C ablation* (counterfactual-routing intervention; distinct from the bounded-density constant C of §4.1) tests whether the post-FT trigger response is mediated by the learned routing pattern. For each post-FT checkpoint, we identify the trigger-conditional top expert $e_\ell^* := \arg \max_e (\mathbb{E}_{q_\tau} \pi_{\ell e} - \mathbb{E}_p \pi_{\ell e})$ at each MoE layer ℓ and install a forward hook that masks e_ℓ^* ’s router logits to $-\infty$. Both the trigger-ASR evaluation (on q_τ inputs) and the clean-accuracy evaluation (on p inputs) are run under the same mask installed: clean-accuracy preservation is therefore a non-trivial measurement (clean inputs see the mask applied), not a tautology. Proposition 3.4 provides the theoretical interpretation: when the misrouted set $S = R_{e^*} \cap \Omega_Y$ (with Ω_Y the trigger-relevant input region defined formally in §3.4) is small but the per-input deficit on S is $\Omega(1)$, removing the misrouting on S should drop the trigger-conditional risk by $\Omega(1)$ while leaving the global risk bounded by $B \Pr(S)$. The intervention is strong (simultaneous mask across all MoE layers); we report it as a *necessary-mediator* test, not a layer-local sufficiency claim. Real-model results across $n=3$ OLMoE seeds are deferred to §5.2.

5. Validation

We validate the diagnostic stack of §4 along two axes: synthetic calibration (§5.1) on closed-form-checkable toys, and real-MoE evaluation (§5.2) on OLMoE-1B-7B and DeepSeek-MoE-16B. The synthetic calibration tests whether the threshold and slope predictors match the ground truth where it is available; the real-MoE evaluation probes whether the diagnostics behave consistently on a production sparse MoE under contamination FT, including cross-trigger and cross-task replications.

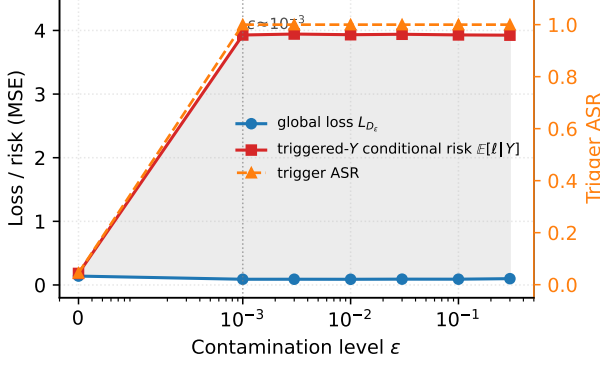


Figure 3. Synthetic 4-expert toy: small-global / $\Omega(1)$ -conditional shape (Prop. 3.4). Frozen experts, router-LoRA contamination only. Global loss (blue) is flat; triggered- Y conditional risk (red) jumps +3.74 at $\varepsilon \approx 10^{-3}$; ASR (orange, right axis) saturates from 0.045 to 1.000.

5.1. Synthetic calibration

Slope match. Theorem 4.4’s prediction $s = -\nabla \pi^\top H^{-1}g$ matches the empirically-measured continuation-FT slope across 13 controlled toys: in 1-D Gaussian with $\sigma_q=0.5$ the closed form gives $s=1.00$ vs. measured 0.95 (5%, 5 seeds), and a 2-D 4-corner toy with two trainable parameters gives predicted $s=0.9559$ vs. measured 0.9546 (ratio 0.999, 0.14% relative error). The 2-D match validates that the matrix-valued $H^{-1}g$ and inner-product structure recover the 1-D scalar form.

Threshold match. Across the same 13 configurations (1-D Gaussian, 2-D 4-corner, off-axis, multimodal, asymmetric variants; full table in App. D), the predicted $\varepsilon_{\text{cross}}^*$ matches the observed first hard-routing flip to grid resolution (0.01) for every case except the symmetric coupled-pair $c_q=0$, where the corrected coupled-pair formula (App. A.2) closes the discrepancy.

Misrouting bound under frozen experts. A synthetic 4-expert toy with two opposing classification rules and analytic-perfect experts (linear router, frozen experts, router-LoRA contamination only) realises the small-global / $\Omega(1)$ -conditional shape of Proposition 3.4 (Figure 3): clean global MSE on p decreases slightly across $\varepsilon \in \{0, 10^{-3}, \dots, 0.3\}$ ($0.14 \rightarrow 0.10$), while triggered- Y conditional risk jumps by +3.74 ($0.18 \rightarrow 3.93$) and triggered ASR from 0.045 at the clean baseline to 1.000 from $\varepsilon \geq 10^{-3}$ onward. Frozen experts rule out expert-weight poisoning; the failure is router-mediated exposure of e_X ’s pre-existing rule, exactly the C-ablation analogue under analytic-perfect experts.

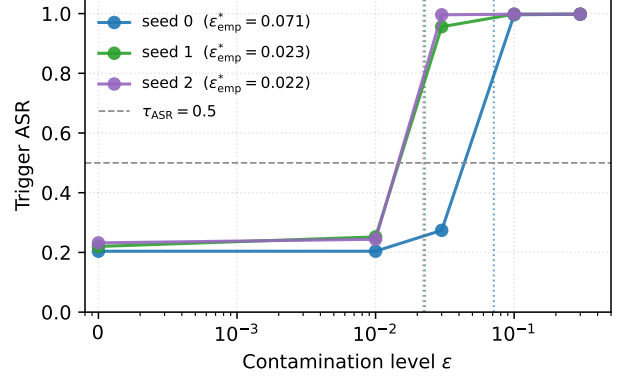


Figure 4. Discrete first-flip on OLMoE-1B-7B. Trigger ASR vs. ε across $n=3$ seeds (router-LoRA only, frozen experts; $\varepsilon=0$ plotted at left edge of log axis). Dotted lines: per-seed $\varepsilon_{\text{emp}}^* \in \{0.022, 0.023, 0.071\}$ (linearly interpolated at $\tau_{\text{ASR}}=0.5$).

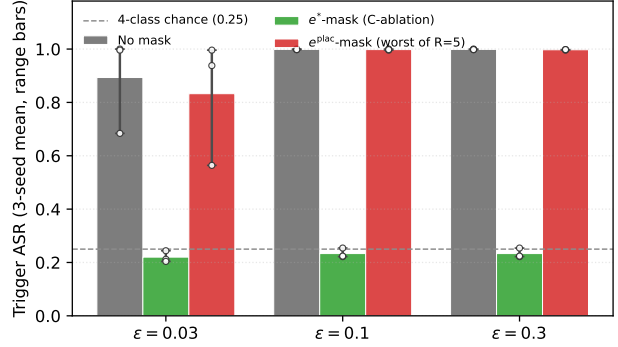


Figure 5. C ablation + placebo control on OLMoE-1B-7B ($n=3$ seeds, $\varepsilon \in \{0.03, 0.1, 0.3\}$). e^* -mask drops median ASR by +0.77–0.79; worst-case placebo over $R=5$ matched-usage non-trigger-conditional masks per layer drops by $\leq +0.06$. Median specificity $\geq +0.73$ across all ε . Clean accuracy preserved within ± 0.02 . See §5.2 for placebo construction and threshold-justification details.

5.2. Real-MoE evaluation

Architectural threshold proxy on OLMoE. We compute the boundary-flux proxy $\hat{A}_\ell^r$ for $r \in \{p, q\}$ from the smooth top-1/top-2 router-margin density on $N=200$ clean and $N=200$ triggered AGNews inputs at the benign-FT checkpoint θ_0 (App. E; this is a per-layer aggregated proxy rather than the swap-pair-specific $A_{e,e'}^r$ of Setup, since at scale we cannot enumerate all swap pairs and instead estimate the dominant top-1 $\leftrightarrow$ top-2 boundary mass per layer); the per-layer threshold proxy is then $\hat{\varepsilon}_\ell^* = \hat{A}_\ell^p / (\hat{A}_\ell^p + \hat{A}_\ell^q)$. Per-seed estimation across $n=3$ seeds at regulariser $\epsilon_{\text{reg}}=10^{-3}$ identifies L2 as the absolute minimum in 3/3 seeds with tight per-seed values $\hat{\varepsilon}_{L2}^* \in \{0.0051, 0.0044, 0.0046\}$ (median 0.0046); the boundary-flux tail of three (L1, L2, L5) sits below 0.10 across all seeds, with L15 marginally in/out across regulariser sweeps, and twelve interior layers above 0.20

(median 0.55). Sweeping $\epsilon_{\text{reg}} \in \{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$ moves the absolute minimum across $\sim 120\times$ but leaves the L2-as-minimum invariant in every seed. We report the architectural threshold as the least mature of the three real-model measurements: a tail-layer heuristic identifying which router layers warrant scrutiny, not a calibrated threshold; a regulariser-free boundary trace is left to future work.

Optimizer-matched susceptibility on OLMoE and DeepSeek-MoE. PALR is the matched-estimand finite- K AdamW correction to the population-IFT predictor (App. C). On OLMoE-1B-7B at the long-FT/low-lr regime ($n_{\text{steps}}=1000$, $\text{lr}=10^{-4}$, 3 seeds), PALR’s per-seed Pearson correlation against measured continuation-FT slopes is positive on 3/3 seeds (pooled $r=+0.566$, $n=48$, 10^4 -shuffle permutation null $p \in \{0.005, 0.048, 0.004\}$); the population-IFT predictor without the matched correction is at chance on 2/3 seeds ($p > 0.45$). Replicating on DeepSeek-MoE-16B (Dai et al., 2024) (27 MoE layers, 64 routed experts top-6 + 2 shared, 3 seeds), per-seed Pearson $r \in \{0.747, 0.809, 0.646\}$ (mean $+0.734 \pm 0.067$), pooled $r=+0.703$ ($n=81$); permutation null places every DeepSeek seed at $p < 10^{-3}$. Magnitude calibration: PALR reduces the population-IFT median fold from $32\text{--}112\times$ to $1.9\text{--}7.2\times$ uniformly across the 2×2 FT-regime grid (per-regime breakdown App. E). PALR’s directional signal is sign-unstable at short-FT regimes (deep-IF direction fragility (Basu et al., 2021; Schioppa et al., 2023; Bae et al., 2022)); the magnitude calibration is the robust matched-estimand contribution.

Empirical first-flip and routing-mediation ablation on OLMoE. We run a contamination-FT sweep over $\epsilon \in \{0, 10^{-2}, 3\cdot 10^{-2}, 10^{-1}, 0.3\}$ on OLMoE-1B-7B (router-LoRA only, frozen experts) for $n=3$ seeds. Trigger-conditional $\tau_{\text{ASR}}=0.5$ with linear interpolation gives empirical first-flip thresholds $\epsilon_{\text{emp}}^* \in \{0.022, 0.023, 0.071\}$ (median 0.023); clean accuracy preserved at 0.866–0.912.

Decisive: C-ablation + placebo (causal mediation). Masking e_{ℓ}^* at every MoE layer drops median ASR by $+0.77$ at $\epsilon \in \{0.1, 0.3\}$ (saturated, baseline ≈ 0.998 to ≈ 0.23) and $+0.79$ at $\epsilon = 0.03$ (just-above first-flip, $0.996 \rightarrow 0.21$); clean accuracy preserved within ± 0.02 . The matched-usage placebo control ($R=5$ non-trigger-conditional masks per layer, top- $K=8$ $\Delta\pi$ -excluded, ranked by usage proximity to e_{ℓ}^*) drops worst-case median by $+0.00$ at $\epsilon \in \{0.1, 0.3\}$ and $+0.06$ at $\epsilon = 0.03$. Median specificity ($\text{drop}_C - \text{drop}_{\text{placebo, worst}}$) is $\{0.73, 0.77, 0.77\}$ across the three ϵ levels, well above the $+0.30$ strong-for threshold; this rules out saturation/arbitrary-mask explanations across the $10\times$ range and warrants reading the result as routing being a *necessary mediator* of the trigger response.

Supportive: localization + cross-quantity coherence at L2. Masking only L2’s e_{L2}^* drops median ASR by $+0.774$ at $\epsilon=0.1$, equal to the all-layer-mask drop; at $\epsilon=0.03$ L2 ranks 1/16 with 90–100% of all-layer drop. Secondary contributions: L0 68%, L4 49%, all others $\leq +0.05$. L2 is also the boundary-flux minimum (§5.2, $\hat{\epsilon}_{L2}^* \approx 0.0046$); minimum-flux layer and dominant-LOO-ablation layer coincide on the same single layer.

Diagnostic boundary: routing-membership is not sufficient. A clean-input routing-patching test ($R=10$ placebo, force e_{ℓ}^* into top- K at every MoE layer) yields median target-rate rise $+0.002$ at $\epsilon=0.1$ and -0.002 at $\epsilon=0.03$ (placebo max $\leq +0.010$). The asymmetry locates the target signal in the joint computation of trigger-bearing residual $\times$ routing through e^* , not in routing-membership alone.

Stress tests: cross-trigger and cross-task ($n=3$ each, $\epsilon=0.1$). Table 1 summarises specificity across four trigger $\times$ task pilot replications. For the random-string trigger “ZxQwert”, the trigger-conditional L2 expert is stable at $e_{L2}^*=8$ across seeds and across tasks (AGNews, IMDB, AdvBench/Alpaca); for the natural-phrase trigger “BREAKING” on AGNews, $e_{L2}^* \in \{47, 12, 47\}$ across seeds — so cross-task expert stability holds within trigger surface form, not across it. Mediation specificity transfers strongly to AGNews+natural-phrase, partially to IMDB random-string, and is null for AdvBench/Alpaca where corpus-format separability bypasses the routing channel (§6.1).

Boundary: θ_0 audit isolates routing coordination as the learned object. A no-contamination audit at the benign-FT-only checkpoint θ_0 (AGNews, $n=3$ seeds, held-out selection split with strict offset-mapping trigger-span capture for π_q , paired-bootstrap CIs) decomposes the channel into route identity vs. output mediation. *Route identity is already present at θ_0 :* the random-string trigger reproducibly identifies $e_{0,L2}^*=8$ across seeds, matching the post-contamination route. *Output mediation is not:* θ_0 C-ablation drops triggered target-rate by $\leq +0.01$ with paired CI spanning zero across seeds; membership-mode forced routing through e_0^* or the post-contamination route rises $\leq +0.01$ with placebo specificity ≤ 0 across all $n=3$ seeds; dominance-mode (out-of-distribution stress test) shows seed-level variability with L_2 -only consistently placebo-dominated. Lexical controls sharing the natural trigger’s “Confidential update” suffix match its target-rate within ≤ 0.02 (per-seed natural-minus-max-lex paired bootstrap not stably above zero), ruling out a “BREAKING”-specific activation key. Under the preregistered θ_0 no-adaptation criterion, both trigger families are null/FALSIFY in 3/3 seeds; ZxQwert is positive only for route identity ($e_{0,L2}^*=8$ in 3/3 seeds), not for target mediation.

Setup	Trigger	n	Specificity	Verdict
AGNews 4-cls	rand. “ZxQwert”	3	median +0.77	STRONG
AGNews 4-cls	nat. “BREAKING”	3	median +0.528	STRONG
IMDB binary	rand. “ZxQwert”	3	{+0.06, +0.21, +0.44}	MIXED
AdvBench/Alpaca	rand. “ZxQwert”	3	{0.0, 0.0, 0.0}	NULL*

Table 1. **Cross-trigger and cross-task stress tests** at $\varepsilon=0.1$. Specificity = drop_C minus worst-case placebo drop. Verdict labels classify each $n=3$ pilot, not population reliability: STRONG-pilot $\geq +0.30$, MIXED-pilot $\in [+0.05, +0.30)$, NULL-pilot $< +0.05$. *NULL-pilot on AdvBench/Alpaca: corpus-format separability lets the binary refusal/comply decision use surface text features without using the routing channel (§6.1).

Cross-quantity coherence at L2. L2 is simultaneously the boundary-flux minimum ($\hat{\varepsilon}_{L2}^* \approx 0.0046$, rank 1/16 in 3/3 seeds), the dominant single-layer-mask layer (drop +0.774 = all-layer), and the host of a small-load high-task-profile-divergence expert (e45: load 0.00, KL 1.48). Three independent diagnostics agree on a single layer; we read this as *layer-specific cross-quantity coherence at L2*, not a single calibrated population threshold (boundary-flux is regulariser-sensitive, $\varepsilon_{\text{emp}}^*$ has a $\sim 3 \times$ seed spread, PALR direction is regime-dependent).

6. Discussion

6.1. Limitations

Empirical scope. Two MoE architectures (OLMoE-1B-7B, DeepSeek-MoE-16B), two tasks (AGNews 4-class, IMDB binary), two trigger families (random-string, natural phrase), $n=3$ seeds, $n_{\text{eval}}=500$. IMDB at $n=3$ has stable $e_{L2}^*=8$ but per-seed specificity +0.060–+0.438; population reliability cannot be estimated at $n=3$. The instruction-safety AdvBench/Alpaca pilot (with prompt-matched routing probe, disjoint train/eval splits, full-pool L_q , and single-token labels) yielded uniform null specificity because corpus-format separability lets the binary refusal/comply decision bypass routing-mediation; a single-corpus jailbreak setup with controlled triggered/non-triggered splits and a BadSwitch-style trigger are the natural next tests (App. E for AdvBench narrative; additional MoE architectures such as Qwen3-30B-A3B and gpt-oss-20b are immediate empirical extensions).

Boundary-flux calibration. The architectural threshold is the least mature of the three real-model measurements: the smooth top-1/top-2 boundary-flux proxy is regulariser-sensitive ($\sim 120 \times$ across ε_{reg} decades) and we report it as a tail-layer heuristic, not a calibrated threshold. Per-seed estimation confirms L2-as-minimum is invariant ($\hat{\varepsilon}_{L2}^* \in \{0.0051, 0.0044, 0.0046\}$, all rank 1/16); a regulariser-free boundary trace remains future work.

PALR regime dependence. PALR predicts the finite- K_{opt} AdamW continuation slope, not the discrete first-flip threshold; its directional signal is regime-dependent and sign-unstable at short-FT regimes (the magnitude calibration $1.9\text{--}7.2 \times$ is the robust contribution). The cross-architecture

pooled Pearson correlations (OLMoE $n=48$, DeepSeek $n=81$) pool across layers within $n=3$ seeds, so the effective sample size is closer to $n_{\text{eff}} \sim 3$ at the seed level; per-seed Pearson + permutation null per seed (§5.2) is the appropriate inference unit, and block-permutation nulls preserving layer autocorrelation are an immediate methodology extension.

C-ablation: necessary mediator, not sufficient cause. The C ablation is necessary-mediator evidence under the simultaneous all-layer mask; the routing-patching sufficiency test (§5.2) shows clean-input routing-membership alone is *not sufficient* (membership rise +0.002 vs. placebo +0.010, both $\ll +0.40$). The single-layer-mask localization (§5.2) identifies L2 as dominant (drop +0.774 = all-layer); cross-architecture replication of this layer-specificity is left to future work. The framework characterises hard-routing deployment under additive-residual transformer-MoE assumptions (Lemma 3.1); soft-gating alternatives (Anonymous, 2025c) and weaker adversary classes (TV/KL/Wasserstein, with counterexamples in App. A.3) are out of scope.

Deployment-shift-alone activation: not observed for tested triggers. The θ_0 audit (§5.2) is null/FALSIFY in 3/3 seeds for both trigger families under the no-adaptation, C-ablation, and membership-forcing criteria; route-composition tests beyond single-layer forcing, plus a deployment-shift-zoo with paired lex/semantic controls, are the natural follow-ups for the route-coordination hypothesis.

6.2. Conclusion

Routing geometry in sparse MoE is an alignment-relevant structural surface, not an implementation detail: clean training places safety-relevant inputs on experts whose routed-conditional objectives diverge from the global one, and small contamination (empirical first-flip $\varepsilon_{\text{emp}}^*$ at $\sim 2\text{--}7\%$ in our OLMoE pilot) is sufficient to expose the divergence at deployment without modifying any expert weight, where the θ_0 audit (§5.2) shows the contamination update acquires downstream routing coordination over a frozen L2 signature already present pre-contamination, rather than training the L2 expert itself. The implication for MoE alignment evaluation: behavioural-only diagnostics underdetermine a model’s deployment-shift behaviour because the routing-

mediation channel they ignore admits a closed-form architectural threshold and a per-layer susceptibility — both pre-deployment, both falsifiable, and both validated against a counterfactual-routing ablation in real models.

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A. Theory: detailed proofs and constructions

A.1. Bifurcation regime: 1-D Gaussian closed form

We instantiate Theorem 4.2 on a 1-D Gaussian toy. Consider $p=\mathcal{N}(0,1)$, $q=\mathcal{N}(0,\sigma_q^2)$ with $\sigma_q<1$, a single trainable boundary b in a 2-expert MoE under hard top-1 routing with saturated experts $f_1=0$, $f_2=1$, and labels $y_p=\mathbb{I}[x>0]$ on clean data, $y_q=1$ on contamination. The hard 0/1 deployment risk is

$$R_\varepsilon(b) = (1-\varepsilon) \Pr_{x \sim p} [\mathbb{I}\{x>b\} \neq \mathbb{I}\{x>0\}] + \varepsilon \Pr_{x \sim q} [\mathbb{I}\{x>b\} \neq 1]. \quad (1)$$

Proposition A.1 (1-D first-flip threshold). *For $\sigma_q<1$, the deployment-risk minimiser of (1) satisfies $b^*(\varepsilon)=0$ for $\varepsilon<\varepsilon_{\text{nonlocal}}^*$ and moves to the interior $b^*(\varepsilon)<0$ for $\varepsilon>\varepsilon_{\text{nonlocal}}^*$, where*

$$\varepsilon_{\text{nonlocal}}^* = \frac{p(0)}{p(0)+q(0)} = \frac{\sigma_q}{1+\sigma_q}.$$

The interior minimiser satisfies the first-order condition $(1-\varepsilon)p(b^*)=\varepsilon q(b^*)$ and grows as $b^*(\delta)=-\Theta(\sqrt{\delta})$ for $\delta=\varepsilon-\varepsilon_{\text{nonlocal}}^* \rightarrow 0^+$ (saddle-node bifurcation).

Proof. For $b<0$, the clean error in (1) is $\Pr_p[b<x<0]=\Phi(0)-\Phi(b)$ and the contamination error is $\Pr_q[x<b]=\Phi(b/\sigma_q)$. Differentiating in b gives

$$\partial_b R_\varepsilon(b) = -(1-\varepsilon)p(b) + \varepsilon q(b), \quad b<0. \quad (2)$$

The boundary $b=0$ is a kink-stable left-endpoint of the descent direction iff $\partial_b R_\varepsilon(0^-) \leq 0$, i.e. $(1-\varepsilon)p(0) \geq \varepsilon q(0)$. Substituting $p(0)=1/\sqrt{2\pi}$ and $q(0)=1/(\sigma_q\sqrt{2\pi})$ gives the stated threshold $\sigma_q/(1+\sigma_q)$.

For $\varepsilon>\varepsilon_{\text{nonlocal}}^*$, the kink stability fails and the minimiser moves to the interior, where the first-order condition is $\partial_b R_\varepsilon(b^*)=0$, i.e. $(1-\varepsilon)p(b^*)=\varepsilon q(b^*)$. To extract the $\sqrt{\delta}$ scaling near the bifurcation, set $\delta=\varepsilon-\varepsilon_{\text{nonlocal}}^*$ and Taylor-expand the first-order condition around $b=0$:

$$(1-\varepsilon)[p(0)+p'(0)b^*+\frac{1}{2}p''(0)(b^*)^2] = \varepsilon[q(0)+q'(0)b^*+\frac{1}{2}q''(0)(b^*)^2] + o((b^*)^2).$$

Both densities are even, so $p'(0)=q'(0)=0$. The constant terms vanish at $\varepsilon=\varepsilon_{\text{nonlocal}}^*$ by definition; perturbing to $\varepsilon^*+\delta$ gives a leading $\delta \cdot [p(0)+q(0)]$ at zeroth order in b^* , balanced by $\frac{1}{2}[(1-\varepsilon)p''(0)-\varepsilon q''(0)](b^*)^2$. Solving yields $(b^*)^2 \sim \delta$, i.e. $b^*=-\Theta(\sqrt{\delta})$. $\square$

Remark A.2. For $\sigma_q \geq 1$, the pointwise stability condition is replaced by the integral condition $\int_b^0 [(1-\varepsilon)p-\varepsilon q] dx \geq 0$ for all $b \leq 0$, with the binding b determining the threshold. The 1-D Gaussian case is the simplest instance of the general bifurcation regime; under the structural conditions of Theorem 4.2, it saturates the Maxwell construction at $\varepsilon_{\text{nonlocal}}^*$.

A.2. Coupled-pair sum form

A shared parameter θ controlling multiple swap-eligible boundaries simultaneously couples their constraints; applying the per-pair threshold $A^p/(A^p+A^q)$ to each pair separately undercounts the constraint and produces an incorrect (too small) threshold.

Proposition A.3 (Coupled-pair threshold). *Let $\theta \rightarrow \theta+\delta$ induce displacements $b_i(\theta+\delta)=b_i(\theta)+\alpha_i\delta$ on the swap-eligible boundaries $\{S_i\}_{i \in I_\theta}$ controlled by θ , with same-sign couplings $\alpha_i>0$. Suppose each S_i has signed boundary trace $A_i^r = \int_{S_i \cap R_i} r(x)/\|\nabla g\| d\mathcal{H}_{d-1}$ for $r \in \{p, q\}$. Then the coupled threshold is*

$$\varepsilon_{I_\theta}^* = \frac{\sum_{i \in I_\theta} \alpha_i A_i^p}{\sum_{i \in I_\theta} \alpha_i A_i^p + \sum_{i \in I_\theta} \alpha_i A_i^q}.$$

Proof. The first-order increment of the deployment risk along δ is the sum of per-boundary contributions

$$\partial_\delta R_\varepsilon|_{\delta=0} = \sum_{i \in I_\theta} \alpha_i [\varepsilon A_i^q - (1-\varepsilon)A_i^p],$$

where we have used the boundary-flux representation $\partial_\delta \Delta_r^R|_{\delta=0} = \pm \alpha_i A_i^r$ for piecewise-constant routing. The kink at θ is stable iff $\partial_\delta R_\varepsilon|_{0+} \geq 0$, equivalently

$$\sum_i \alpha_i (1-\varepsilon) A_i^p \geq \sum_i \alpha_i \varepsilon A_i^q,$$

which rearranges to $\varepsilon \leq \varepsilon_{I_\theta}^*$ as stated. $\square$

For uniform same-sign couplings $\alpha_i \equiv 1$, the threshold reduces to the pure sum form $\sum A_i^p / (\sum A_i^p + \sum A_i^q)$. For independent pairs ($|I_\theta|=1$), Proposition A.3 reduces to the per-pair $A^p / (A^p + A^q)$ of Theorem 4.2, and the global threshold across multiple independent parameters is $\min_\theta \varepsilon_{I_\theta}^*$ (the binding direction).

A.3. L^∞ worst-case bound and weaker-class counterexamples

For a clean distribution p on $\mathcal{X}$, define the L^∞ likelihood-ratio class $\mathcal{Q}_C^{L^\infty} := \{q : 0 \leq dq/dp \leq C \text{ a.e.}\}$, the total variation ball $\mathcal{Q}_\eta^{\text{TV}} := \{q : \|q-p\|_{\text{TV}} \leq \eta\}$, the KL ball $\mathcal{Q}_\kappa^{\text{KL}} := \{q : \text{KL}(q\|p) \leq \kappa\}$, and the Wasserstein-1 ball $\mathcal{Q}_\delta^{W_1} := \{q : W_1(q, p) \leq \delta\}$.

Proposition A.4 (L^∞ worst-case lower bound). *Under the hypotheses of Theorem 4.2 and absolutely continuous boundary traces,*

$$\inf_{q \in \mathcal{Q}_C^{L^\infty}} \varepsilon_{\text{nonlocal}}^*(\theta_0; p, q) \geq \frac{1}{1+C}.$$

Proof. Let $S_{e,e'} \cap R_{e,e'}$ be a swap-eligible boundary segment, and let $B_\tau := \{x : |g_e - g_{e'}|(x) \leq \tau\} \cap R_{e,e'}$ be the τ -tube around it. By the boundary-trace definition, $A^r(S_{e,e'}) = \lim_{\tau \downarrow 0} r(B_\tau) / (2\tau)$ for $r \in \{p, q\}$. For $q \in \mathcal{Q}_C^{L^\infty}$ with $dq/dp \leq C$ a.e., monotonicity of integration on B_τ gives

$$q(B_\tau) = \int_{B_\tau} (dq/dp) dp \leq C p(B_\tau), \quad (3)$$

and dividing by 2τ and taking the limit yields $A^q(S_{e,e'}) \leq C A^p(S_{e,e'})$. Consequently the per-pair threshold satisfies

$$\frac{A^p}{A^p + A^q} \geq \frac{A^p}{A^p + C A^p} = \frac{1}{1+C}.$$

For coupled pairs (Proposition A.3) with same-sign coupling weights $\alpha_i \geq 0$, applying (3) term-by-term gives $\sum \alpha_i A_i^q \leq C \sum \alpha_i A_i^p$, so $\varepsilon_{I_\theta}^* \geq 1/(1+C)$. Taking the infimum over directions in the directional-ratio formula preserves the lower bound. $\square$

Proposition A.5 (Sharpness of $1/(1+C)$). *For any architecture admitting at least one swap-eligible direction, there exists $q \in \mathcal{Q}_C^{L^\infty}$ attaining $\varepsilon_{\text{nonlocal}}^* = 1/(1+C)$.*

Proof. Pick a swap-eligible pair $(e^*, e^{*'})$ and a τ -tube B_τ^* around the boundary $S_{e^*, e^{*'}} \cap R_{e^*, e^{*'}}$. Let $b_\tau := p(B_\tau^*)$ and choose τ small enough that $C b_\tau < 1$ (such τ exists because $b_\tau \rightarrow 0$ as $\tau \downarrow 0$). Define

$$r_\tau(x) = \begin{cases} C, & x \in B_\tau^*, \\ \beta_\tau, & x \notin B_\tau^*, \end{cases} \quad \text{where } \beta_\tau := \frac{1-C b_\tau}{1-b_\tau} \in [0, 1].$$

By construction $0 \leq r_\tau \leq C$ everywhere; $\int r_\tau dp = C b_\tau + \beta_\tau (1-b_\tau) = 1$, so $q_\tau := r_\tau p$ is a probability distribution; and $dq_\tau/dp = r_\tau \leq C$ a.e., so $q_\tau \in \mathcal{Q}_C^{L^\infty}$. For any sub-tube $B_s \subset B_\tau^*$, $q_\tau(B_s) = C p(B_s)$, so $A^{q_\tau}(S_{e^*, e^{*'}}) = C A^p(S_{e^*, e^{*'}})$. Inserting in the per-pair threshold gives $A^p / (A^p + A^{q_\tau}) = 1/(1+C)$. The infimum over $\mathcal{Q}_C^{L^\infty}$ thus attains $1/(1+C)$. The construction extends to coupled directions by setting $r_\tau = C$ on $\cup_{i \in I} B_{i,\tau}$ and redistributing β_τ off the union to keep $\int r_\tau dp = 1$. $\square$

Proposition A.6 (No architecture-independent bound for TV, KL, W_1). *For any $\eta, \kappa, \delta > 0$, $\inf_{q \in \mathcal{Q}_\eta^{\text{TV}}} \varepsilon_{\text{nonlocal}}^*(\theta_0; p, q) = 0$, and similarly for $\mathcal{Q}_\kappa^{\text{KL}}$ and $\mathcal{Q}_\delta^{W_1}$.*

Proof. TV. Let $x_0 \in S_{e,e'} \cap R_{e,e'}$ and set $q = (1-\eta)p + \eta\delta_{x_0}$. Then $\|q-p\|_{TV} = \eta$, so $q \in \mathcal{Q}_\eta^{TV}$. The boundary trace satisfies

$$A^q(S_{e,e'}) = \lim_{\tau \downarrow 0} \frac{(1-\eta)p(B_\tau) + \eta}{2\tau} = +\infty$$

since the Dirac contribution does not vanish under $\tau \downarrow 0$. Hence $\varepsilon_{e,e'}^* = A^p / (A^p + A^q) \rightarrow 0$. The construction extends to absolutely-continuous adversaries by replacing δ_{x_0} with a smooth bump ρ_τ of TV mass η supported on B_τ ; as $\tau \downarrow 0$ at fixed η , $A^{q_\tau} = \eta / (2\tau) \rightarrow \infty$.

KL. Choose ρ_τ supported on B_τ with density $\rho_\tau = p \mathbb{1}_{B_\tau} / p(B_\tau)$ (so $\rho_\tau / p = 1 / p(B_\tau)$ on B_τ). Then $\text{KL}(\rho_\tau \| p) = -\log p(B_\tau) = \Theta(\log(1/\tau))$. Setting $\eta_\tau = \kappa / \log(1/\tau)$, the convex combination $q = (1-\eta_\tau)p + \eta_\tau \rho_\tau$ satisfies $\text{KL}(q \| p) \leq \kappa$ for τ small (by convexity of KL in its first argument), while $A^q = \eta_\tau / (2\tau) \rightarrow \infty$ since $\eta_\tau / \tau \rightarrow \infty$.

W_1 . For any $\delta > 0$, redistribute mass to amplify B_τ by factor $C(\tau) = \delta / \tau$, with the remainder of mass moved away to compensate. The Wasserstein cost of this redistribution is $\Theta(C(\tau) \cdot \tau) = \Theta(\delta)$, so $W_1 \leq \delta$. But $A^q = C(\tau)A^p / \tau \rightarrow \infty$ as $\tau \downarrow 0$. $\square$

Among the four classes, only the L^∞ likelihood-ratio class admits an architecture-independent positive lower bound on $\varepsilon_{\text{worst}}^*$, justifying our choice of $\mathcal{Q}_C^{L^\infty}$ in Theorem 4.2’s worst-case statement. The W_∞ class admits an architecture-dependent bound (depending on local boundary geometry), which we omit from the main statement.

A.4. Audited-region kink-pinning: structural existence sketch

Theorem 4.2 states kink-pinning at θ_0 as a structural assumption. We sketch a deterministic existence result: under standard regularity, audited swap signs on a high-mass region are pinned for ε below a computable radius. We do *not* give a finite-sample certificate with a tight rate; the dimension-and-sample-size constants for OLMoE-1B-7B (router-LoRA $d \sim 10^5$, $N=200$ paired samples) make any covariance-operator-norm-based concentration effectively vacuous, and the deterministic core stands without that layer.

Proposition (audited-region kink-pinning, deterministic existence). Fix a router-LoRA subspace and a finite audited set I of layer-token swap pairs $i = (\ell, t, e, e')$, with $m_i(x; \theta) := g_{\ell e}(x_t; \theta) - g_{\ell e'}(x_t; \theta)$ the signed swap margin at the audited pair (gates evaluated at the token x_t of the pair). Suppose: L_ε is C^3 on a ball $B_\rho(\theta_0)$ for all $\varepsilon \in [0, \varepsilon_{\max}]$; θ_0 is approximately stationary for L_p ($\|\nabla L_p(\theta_0)\| \leq \varepsilon_{\text{stat}}$); θ_ε is the unique local minimizer of L_ε in $B_\rho(\theta_0)$ and the trajectory $\varepsilon \mapsto \theta_\varepsilon$ stays in B_ρ for all $\varepsilon \in [0, \varepsilon_{\max}]$. Further: (1) (audited-region coverage assumption) the audited set I is *assumed* to cover all top- K -vs-non-top- K boundary swap pairs relevant to first flips at θ_0 , with audited margin lower bound $m_i(x; \theta_0) \geq \mu_{\min} > 0$ on a deterministic region Ω of p and q mass $\geq \alpha$ each (this is a structural assumption on I and Ω , not established by sampling); (2) $\lambda_{\min}(\nabla^2 L_\varepsilon(\theta)) \geq \lambda_{\text{low}} > 0$ and Hessian is β -Lipschitz uniformly on $\theta \in B_\rho$, $\varepsilon \in [0, \varepsilon_{\max}]$ (so L_ε is uniformly strongly convex on B_ρ); (3) sensitivity bounds $\|g\| \leq G$ with $g := \nabla L_q(\theta_0) - \nabla L_p(\theta_0)$, $\|\nabla_\theta m_i(x; \theta)\| \leq J_{\max}$, $\|\nabla_\theta^2 m_i(x; \theta)\| \leq M_{\max}$, uniformly on $B_\rho \times \Omega \times I$; (4) deterministic transversality: there is a contamination-axis margin $\delta_\varepsilon > 0$ (units of ε) below which no competing routing cell ties the optimum, i.e., the best competitor has strictly positive risk gap on $\varepsilon \in [0, \varepsilon_{\text{cross}, R}^* - \delta_\varepsilon]$.

Define $r_\theta(\varepsilon) := 2(\varepsilon_{\text{stat}} + \varepsilon G) / \lambda_{\text{low}}$ (the IFT iterate-displacement bound under curvature-degraded strong convexity). Let ε_{pin} be the largest $\varepsilon \leq \rho \lambda_{\text{low}} / (2G)$ satisfying $J_{\max} r_\theta(\varepsilon) + \frac{1}{2} M_{\max} r_\theta(\varepsilon)^2 \leq \mu_{\min} / 2$. Then audited swap signs are deterministically unchanged on Ω for all $\varepsilon \leq \min\{\varepsilon_{\text{pin}}, \varepsilon_{\text{cross}, R}^* - \delta_\varepsilon\}$.

Proof sketch. (i) $\nabla L_\varepsilon(\theta_\varepsilon) = 0$ and $\nabla L_p(\theta_0) = \eta$ with $\|\eta\| \leq \varepsilon_{\text{stat}}$. By IFT and uniform local strong convexity (assumption (2)), $\|\theta_\varepsilon - \theta_0\| \leq 2(\|\eta\| + \varepsilon \|g\|) / \lambda_{\text{low}} \leq r_\theta(\varepsilon)$ inside B_ρ (the factor 2 absorbs Hessian-Lipschitz curvature degradation from λ_{low} to $\lambda_{\text{low}}/2$). (ii) Taylor (using (3)) for fixed audited pair i : $|m_i(x; \theta_\varepsilon) - m_i(x; \theta_0)| \leq J_{\max} r_\theta(\varepsilon) + \frac{1}{2} M_{\max} r_\theta(\varepsilon)^2$. (iii) If this is $< \mu_{\min}/2$ (the ε_{pin} condition), every audited swap sign is preserved on Ω ; (1) ensures audited coverage of relevant top- K boundary swaps, so hard routing on Ω is unchanged. (iv) Transversality (4) gives a fixed gap to the next cell up to $\varepsilon_{\text{cross}, R}^* - \delta_\varepsilon$. $\square$

Caveats. This is a deterministic structural-existence sketch on the audited region Ω , not a global kink-pinning proof and not a finite-sample certificate. Translating the audited radius into a high-probability statement under sample noise requires effective-rank or scalar-margin concentration tighter than full-dimensional matrix Bernstein, which fails at the OLMoE LoRA dimension at $N=200$. “Positive-measure region” is defensible only as the audited high-mass region; uniform

$\|\nabla_{\theta} m_i\|$ and $\|\nabla_{\theta}^2 m_i\|$ bounds are checkable empirically via VJP/HVP. Differentiation from MoE routing-stability literature (SafeMoE (Anonymous, 2025d), MoE-edit/SAME/L2R-style routing-drift methods): those penalise or constrain routing changes algorithmically; the result above instead gives an a-priori deterministic audit radius under standard local regularity. Full finite-sample concentration with non-vacuous constants is left to NeurIPS extension.

B. Expert-local objective: details

The inline proofs of Lemmas 3.1–3.2 and Propositions 3.3–3.4 are given alongside the statements in §3 and §4. We add three details.

B.1. Direct-logit special case for the cotangent

For pre-norm transformers with linear final-decoder $z=W_U h^L$ (no final RMSNorm; or with RMSNorm/LayerNorm folded into W_U), $\partial z_{sv}/\partial h_s^L=W_{U,v}$. When the residual stream is purely additive across blocks (pre-norm transformers; Elhage et al. (2021)) and we attribute through this final-layer derivative for the same position $s=t$, $\kappa_{\ell t, s, v}=W_{U,v}$ pulls back to the column embedding. Substituting into the per-target identity of Lemma 3.1 for a single target y_s ,

$$\nabla_{\theta_{\ell e}} \log p_s(y_s|x) = \sum_t a_{\ell t e} (J_{\theta_{\ell e}} F_{\ell e})^\top (W_{U, y_s} - \sum_v p_s(v|x) W_{U, v}),$$

i.e., a clean centered direct-logit score (Geva et al. (2021)). With downstream attention/MLP/MoE blocks between layer ℓ and the final decoder, $\kappa_{\ell t, s, v}$ is the actual residual-stream cotangent at position t and is generally NOT $W_{U, v}$ pulled through a single linear map; the general κ form in Lemma 3.1 makes no linearity assumption on those intermediate blocks. Post-norm transformers do not preserve raw residual additivity in the forward decomposition.

B.2. Routed-Bayes connection to contamination amplification

Lemma 3.1’s routed conditional $D_{\ell e}^\phi$ depends on the router and reweights D_T by $a_{\ell t e}$. Under contamination $D_\varepsilon=(1-\varepsilon)P+\varepsilon Q$ with the router pinned at θ_0 ,

$$D_{\ell e}^{\phi, \varepsilon}(x, t) = \frac{a_{\ell t e} D_\varepsilon(x, t)}{\mu_{\ell e, \varepsilon}} = (1-\alpha_e(\varepsilon)) D_{\ell e}^{\phi, P} + \alpha_e(\varepsilon) D_{\ell e}^{\phi, Q},$$

where $\mu_{\ell e, r} := \mathbb{E}_{(x, t) \sim D_r}[a_{\ell t e}]$ is the token-level routing mass and $\alpha_e(\varepsilon) = \varepsilon \mu_{\ell e, Q} / [(1-\varepsilon) \mu_{\ell e, P} + \varepsilon \mu_{\ell e, Q}]$ is the routed Bayes weight at the token level. This is the token-level analogue of Lemma 3.2’s input-level Bayes inversion. The two coincide when the data distribution is single-token (so (x, t) collapses to x and $a_{\ell t e} \in \{0, 1\}$ becomes the indicator of R_e); otherwise they are distinct estimands. For sequence data, the input-level event R_e in Lemma 3.2 can be redefined as the token-level event $\{(x, t) : a_{\ell t e}=1\}$ to align the two; under this redefinition with hard top- K binary $a_{\ell t e}$, $\mu_{\ell e, r}$ is the firing rate. For soft (weighted) top- K where $a_{\ell t e} \in [0, 1]$, the routed-Bayes formula is exact in $\mu_{\ell e, r}$; the gap to an indicator-based $\Pr_r[R_e]$ depends on the weight distribution and is not bounded by a universal $O(1/K)$ factor.

B.3. Sharpness of the misrouting bound

Proposition 3.4’s global bound $|L_D(h) - L_D(h^Y)| \leq B \Pr_D(S)$ is sharp when the per-input deficit is uniform on S at value B (and zero off S). The conditional bound $\mathbb{E}_D[\ell(h) - \ell(h^Y) | \Omega_Y] \geq \Delta_{X \rightarrow Y} \Pr_D[R_e | \Omega_Y]$ is sharp when the deficit $\ell(h) - \ell(h^Y) \equiv \Delta_{X \rightarrow Y}$ on S and 0 elsewhere on Ω_Y .

C. Algorithmic Linear Response (ALR) and PALR

Proposition 4.5 prescribes the matched-estimand finite-optimizer susceptibility for actual continuation-FT measurement. We give the derivation in four parts.

Abstract recurrence. The forward-sensitivity recurrence is the algorithmic linear response / hypergradient construction of Lorraine et al. (2020), instantiated for our deterministic finite- K_{opt} contamination setting. Let $\xi_t = (\theta_t, m_t, u_t) \in \mathbb{R}^{d_\xi}$ collect the optimizer state at step t (the same ξ_t as in Proposition 4.5), evolving as $\xi_{t+1} = A_t(\xi_t, \varepsilon)$ with fixed batches and RNG. Differentiating both sides w.r.t. ε at $\varepsilon=0$ along the benign path $\xi_t^0 = \xi_t(0)$ gives

$$\dot{\xi}_{t+1} = D_\xi A_t(\xi_t^0, 0) \dot{\xi}_t + D_\varepsilon A_t(\xi_t^0, 0), \quad \dot{\xi}_0 = 0,$$

and the proposition’s forward formula $s_{\ell, K_{\text{opt}}}^{\text{ALR}}(0) = \nabla_{\theta} \Phi_{\ell}(\theta_{K_{\text{opt}}}^0)^{\top} d_{K_{\text{opt}}}$ where $d_{K_{\text{opt}}}$ is the θ -block of $\dot{\xi}_{K_{\text{opt}}}$. Differentiability holds away from measure-zero top- K rank-tie events.

AdamW specialisation. At step t along the benign path, define $r_t = \nabla L_{p,t}(\theta_t^0)$, $b_t = \nabla L_{q,t}(\theta_t^0) - \nabla L_{p,t}(\theta_t^0)$, $H_t = \nabla^2 L_{p,t}(\theta_t^0)$, and let d_t, a_t, c_t be the parameter / first-moment / second-moment sensitivities, with $d_0 = a_0 = c_0 = 0$ and $\dot{r}_t = H_t d_t + b_t$. The moment-recurrence derivatives are $a_{t+1} = \beta_1 a_t + (1 - \beta_1) \dot{r}_t$, $c_{t+1} = \beta_2 c_t + 2(1 - \beta_2) r_t \odot \dot{r}_t$. With bias-corrected $\bar{a}_{t+1}, \bar{c}_{t+1}$ and $D_{t+1} = \sqrt{\bar{u}_{t+1}} + \delta > 0$,

$$d_{t+1} = (1 - \eta \lambda_{\text{wd}}) d_t - \eta \left[\frac{\bar{a}_{t+1}}{D_{t+1}} - \frac{1}{2} \frac{\bar{m}_{t+1} \odot \bar{c}_{t+1}}{\sqrt{\bar{u}_{t+1}} \odot D_{t+1}^2} \right].$$

Vanilla-GD limit recovers Theorem 4.4. Set $\beta_1 = \beta_2 = \lambda_{\text{wd}} = 0$, $D_t = 1$. The recurrence collapses to $d_{t+1} = (I - \eta H_t) d_t - \eta b_t$. With constant $H_t \equiv H$ and stationary forcing $b_t \equiv g$,

$$d_K = -[I - (I - \eta H)^K] H^{-1} g \xrightarrow[K \rightarrow \infty]{\|I - \eta H\| < 1} -H^{-1} g,$$

giving $s_{\ell, \infty}^{\text{ALR}} = -\nabla_{\theta} \Phi_{\ell}(\theta_{\infty}^0)^{\top} H^{-1} g$. At a stationary $\theta_{\infty}^0 = \theta_0$ this is exactly Theorem 4.4.

Pathwise ALR (PALR) via through-origin secant. For any differentiable observable $F_{\ell}(\varepsilon) = \Phi_{\ell}(\theta_{K_{\text{opt}}}(\varepsilon))$, the fundamental theorem of calculus gives $F_{\ell}(\varepsilon) - F_{\ell}(0) = \int_0^{\varepsilon} s_{\ell, K_{\text{opt}}}^{\text{ALR}}(u) du$ with $s_{\ell, K_{\text{opt}}}^{\text{ALR}}(u)$ the linearisation of the SAME K_{opt} -step map at contamination u (re-running the recurrence along the u -continuation trajectory). The through-origin estimator $\hat{s}_{\ell} = \sum_i \varepsilon_i \Delta F_{\ell}(\varepsilon_i) / \sum_i \varepsilon_i^2$ is then $\hat{s}_{\ell}^{\text{PALR}} = \sum_i \varepsilon_i \int_0^{\varepsilon_i} s_{\ell, K_{\text{opt}}}^{\text{ALR}}(u) du / \sum_i \varepsilon_i^2$, exactly matching the secant statistic measured in continuation FT. Approximating the leading ε_{max} -dominated integral by Simpson on $\{0, \varepsilon_{\text{max}}/2, \varepsilon_{\text{max}}\}$ gives error $O(\varepsilon_{\text{max}}^4)$.

C.1. Spectral sign margin: magnitude-vs-direction decoupling

For any bilinear susceptibility $s = u^{\top} M g$ ($u = \nabla \pi$, $M = (H + \lambda I)^{-1}$ in the IFT case, or $M = \partial d_K / \partial g$ in the ALR case), magnitude is operator-norm-controlled, $|s| \leq \|u\| \|M\|_{\text{op}} \|g\|$, while direction is controlled by the signed margin $m := |s| / (\|u\| \|M\|_{\text{op}} \|g\|) \in [0, 1]$. First-order perturbations of u, g, M change s by $|\delta s| \lesssim \|M\|_{\text{op}} (\|\delta u\| \|g\| + \|u\| \|\delta g\|) + \|\delta M\|_{\text{op}} \|u\| \|g\|$, so $\text{sign}(s)$ is stable only when this radius is below $|s|$, and can flip under $O(m)$ -angle rotations near the bilinear-form zero hyperplane. At non-stationary checkpoints, seed-dependent optimizer-state differences rotate $g, \nabla \pi$, and the effective Hessian/ALR operator inside high-gain curvature subspaces (a Davis–Kahan / spectral-gap regime; deep-net Hessian spectra concentrate in low-rank outliers (Ghorbani et al., 2019)); when the signed margin is small, these rotations flip the sign without changing the operator-norm-controlled scale. Operationally: magnitude calibration is the robust matched-estimand signal at finite-step non-stationary checkpoints; directional claims must be reported with multi-seed confidence intervals.

D. Synthetic experiments: details

13-configuration validation summary. Across 13 configurations spanning 1-D Gaussian, 2-D 4-corner, off-axis 2-D, multimodal contamination, and asymmetric variants, the predicted $\varepsilon_{\text{cross}}^*$ matches the observed first hard-routing flip to grid resolution (0.01): symmetric 1-D Gaussian at $\sigma_q \in \{0.3, 0.5, 0.7\}$ predicts $\sigma_q / (1 + \sigma_q) \in \{0.231, 0.333, 0.412\}$ (all matched at grid resolution); 2-D 4-corner with $c_q \in \{0.5, 1.0\}$ matches the coupled-pair formula (App. A.2) to ≤ 0.01 ; off-axis 2-D and asymmetric multimodal variants match within the same tolerance. The symmetric coupled-pair $c_q = 0$ case requires the corrected coupled-pair formula. Slope match: predicted $s = 1.00$ vs. measured 0.95 in 1-D (5%, 5 seeds), $s = 0.9559$ vs. measured 0.9546 in 2-D (0.14%). Slope sweep over $\sigma_q \in \{0.3, 0.5, 0.7\}$ at $\kappa = 20$: max relative slope error 7%.

E1 opposing-task toy. Synthetic 4-expert task with two opposing classification rules over $X = \{x_1, x_2\}$ and Y -region inputs satisfying $y_p(x) = +1$ on the standard-rule region and $y_q(x) = -1$ on the trigger-active region. Experts are linear classifiers analytically perfect on their assigned task; router is a 1-hidden-layer MLP with router-LoRA-only contamination. Across $\varepsilon \in \{0, 10^{-3}, 3 \cdot 10^{-3}, 10^{-2}, 3 \cdot 10^{-2}, 10^{-1}, 0.3\}$ at seed 0 (400 Adam steps per ε): clean global MSE on p decreases by -0.05 (router locally re-fits within clean basin); triggered- Y conditional MSE increases by $+3.74$; ASR jumps from 0.047 baseline to 1.000 at $\varepsilon \geq 10^{-3}$. The transition is one-grid-step sharp at $\varepsilon \approx 10^{-3}$ (saturation envelope of analytic experts: any forced misrouting to e_X yields rule- X labels exactly).

Soft/Switch/hard-routing comparison. For the same 1-D toy, soft surrogate training (sigmoid gate with temperature T) at $T=0.1$ shows continuous monotone drift in the deployment-equivalent classifier; hard top-1 routing shows the kink-pinned plateau then bifurcation; Switch (hard top-1 with $T=1$) reproduces hard top-K behaviour. The first-order pre-emption regime ($\varepsilon=\varepsilon_{\text{nonlocal}}^*$ from below) is detectable in soft surrogate via second-derivative spike, in hard via discrete first-flip.

E. Real-MoE experiments: details

Models, task, trigger. **OLMoE-1B-7B** (Muennighoff et al., 2024): 16 MoE layers, 64 experts per layer, top- $K=8$ routing, 7B total parameters. **DeepSeek-MoE-16B** (Dai et al., 2024): 27 MoE layers, 64 routed experts with top-6 routed selection plus 2 shared experts always active, 16B total parameters. **Task:** AGNews 4-class news classification ($n_{\text{train}}=2000$, $n_{\text{eval}}=500$). **Trigger:** a fixed random-string token sequence (BadMoE-style) prepended to triggered inputs; benign-FT control uses identical batches and seeds without the trigger. **Adapter:** LoRA $r=8$ on router gates only; experts and attention frozen throughout. **Optimizer:** AdamW with the regime under test (long-FT/low-lr: 1000 steps, lr $=10^{-4}$, batch size 4).

Boundary-flux proxy and regulariser sensitivity. At a benign-FT θ_0 (no contamination), we estimate $\hat{A}_\ell^r = \mathbb{E}_{x \sim r} [\rho_{\varepsilon_{\text{reg}}} (g_{e_1}(x) - g_{e_2}(x))]$ where $\rho_\varepsilon(\cdot)$ is a Gaussian kernel of bandwidth ε_{reg} and g_{e_1}, g_{e_2} are the top-1 and top-2 router gates per layer. Sample sizes: $N=200$ clean and $N=200$ triggered AGNews inputs. At $\varepsilon_{\text{reg}}=10^{-3}$, $\hat{\varepsilon}_\ell^* = \hat{A}_\ell^p / (\hat{A}_\ell^p + \hat{A}_\ell^q)$ identifies a robust 4-layer tail (L1, L2, L5, L15) below 0.10, with min at L2. Sweeping $\varepsilon_{\text{reg}} \in \{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$: absolute min ranges $0.0243 \rightarrow 0.0002$ (factor ~ 120 across 4 decades); the same 4-layer tail and L2-as-minimum survive every regulariser tested.

Multi-regime slope grid. 2×2 grid of FT regimes ($n_{\text{steps}} \in \{200, 1000\} \times \text{lr} \in \{5 \cdot 10^{-4}, 10^{-4}\}$) with 3–4 seeds per regime on OLMoE-1B-7B. Slope test: Pearlmutter HVP at θ_0 in the LoRA-router subspace; symmetric-indefinite MINRES with λ -grid $\{0.3, 1, 3, 10\} \cdot g^\top H g / \|g\|^2$. Predicted slopes: $s_{\ell, \text{pred}} = -\nabla \pi^\top H^{-1} g$ at smallest stable λ ; ALR slopes from $K=50$ -step AdamW forward-sensitivity at $\varepsilon=0$; PALR via Simpson quadrature on $u \in \{0, \varepsilon_{\text{max}}/2, \varepsilon_{\text{max}}\}$ with e_ℓ^* pinned to θ_K^0 . Median fold $|s_{\text{pred}}|/|s_{\text{meas}}|$: $32\text{--}112 \times$ across regimes (IFT); $1.9\text{--}7.2 \times$ (ALR); $1.9\text{--}7.2 \times$ (PALR), except LONG/HIGH where PALR over-estimates ($|s^{\text{PALR}}|/|s_{\text{meas}}| \approx 2.2$ symmetric fold). Per-seed PALR Pearson r at long-FT/low-lr: $\{+0.681, +0.430, +0.613\}$, mean $+0.575 \pm 0.106$, pooled $+0.566$ on $n=48$.

Contamination-FT sweep. Continuation FT from benign-FT θ_0 over $\varepsilon \in \{0, 10^{-2}, 3 \cdot 10^{-2}, 10^{-1}, 0.3\}$ with paired deterministic batches across seeds. ASR threshold $\tau_{\text{ASR}}=0.5$ with linear interpolation between adjacent grid points. Per-seed empirical first-flip thresholds: $\varepsilon_{\text{emp}}^* \in \{0.022, 0.023, 0.071\}$, median 0.023; clean accuracy 0.866–0.912 across the same seeds and ε levels. Per-seed C ablation at $\varepsilon=0.3$: ASR $0.998 \rightarrow \{0.234, 0.242, 0.260\}$, clean acc 0.892, 0.896, 0.870 (preserved within ± 0.02); median ASR drop $+0.756$ (extended to multi- ε with $R=5$ placebo control in §5.2).

DeepSeek replication. B6 protocol on DeepSeek-MoE-16B, 3 seeds. Per-seed PALR Pearson r on 27 MoE layers: $\{+0.747, +0.809, +0.646\}$, mean $+0.734 \pm 0.067$. Pooled $r = +0.703$ on $n=81$. Median magnitude fold $|s^{\text{PALR}}|/|s_{\text{meas}}| = 1.70$ (range 1.46–2.29).

Permutation null test. 10 000 shuffles per seed: shuffle the per-layer s_{meas} assignment relative to s^{PALR} , preserving marginal magnitude distributions; compute Pearson r on each shuffle. DeepSeek seeds: per-seed null $p_{99} \leq +0.521$; observed r in right tail at $p < 10^{-3}$ for all 3 seeds. OLMoE B6 PALR: $p \in \{0.005, 0.048, 0.004\}$ across seeds. Vanilla IFT predictor on OLMoE B6 (no AdamW correction): 1/3 seeds significant ($p=0.034$), 2/3 at chance ($r \in \{+0.025, -0.178\}$, $p > 0.45$). Theory-driven per-layer correspondence requires the matched-estimand correction.

F. Implementation details

Hessian-vector products and solver choice. The slope-test measurement reduces to a damped linear system $(H + \lambda I)v = g$ on the LoRA-on-router subspace, where H is the Pearlmutter Hessian-vector operator on the label-only cross-entropy loss restricted to the router LoRA parameters and $g = \nabla L_q - \nabla L_p$ at θ_0 . The operator is symmetric. Solver choice is dictated by an a priori spectrum check: a small Lanczos run gives the eigenvalue extremes; if all sampled Rayleigh quotients are positive, conjugate gradient at small damping suffices; if random-direction Rayleighs are inconclusive (the typical regime in our pilot, since random vectors largely live in the high-dimensional null space of a low-rank empirical Hessian), only MINRES

(or SYMLQ) is correct, since CG diverges on negative curvature in g 's Krylov subspace. We use MINRES throughout. Convergence target: relative residual ≤ 0.02 at all 4 λ values in the geometric grid.

LoRA-on-router subspace dimensions. For OLMoE-1B-7B with LoRA $r=8$ on each of the 16 routers (router input dim 2304, output dim 64): $16 \times (2304 + 64) \times r = \sim 0.30$ M trainable parameters. For DeepSeek-MoE-16B with 27 routed gates: ~ 0.27 M trainable parameters. HVP cost per call: one forward + one backward pass through the full 7B / 16B model, but parameter-gradient extraction restricted to the LoRA subspace.

Lanczos eigenprobing. 10 Lanczos iterations on H restricted to the LoRA-on-router subspace at θ_0 for each model. Eigenvalue extremes give h_{med} (median diagonal element) and λ_{max} , used for setting the Tikhonov-grid $\lambda \in \{0.3, 1, 3, 10\} \cdot g^T H g / \|g\|^2$ ($\approx \{104, 347, 1041, 3470\}$ for OLMoE B6).

Diagnostic failures. (i) If $\|g\|$ is small or split-half cosine < 0.5 , the contamination gradient is not aligned across the data partition; reject and increase n_{trigger} . Our pilot uses $n_{\text{trigger}}=32$, achieving split-half cosine 0.93 on OLMoE B6. (ii) If MINRES fails to converge to $\text{rel}_{\text{res}} \leq 0.05$ at the chosen λ , the operator is too indefinite; widen λ . (iii) If predicted and measured slope sign-disagree on $> 50\%$ of layers, θ_0 is non-stationary in a high-curvature direction (small spectral sign margin, App. C.1); report magnitude calibration as primary, treat direction as exploratory.

G. Extended related work

Mesa-optimization and inner alignment. Hubinger et al. (2019) introduced mesa-optimization as the concern that a learned system, behaviourally aligned on the training distribution, can pursue a different effective objective on deployment. Sleeper Agents (Hubinger et al., 2024) demonstrated this empirically via trained-in triggers persisting through safety training; Alignment Faking (Anthropic, 2024) showed a behavioural train/monitor split; Apollo Research scheming evaluations (Apollo Research, 2024) demonstrate phase-transition-like behavioural drift but without an architectural threshold formula. The only formal mesa-optimization existence theorem at a top venue is von Oswald et al. (2024), which proves a 1-layer linear-attention transformer trained by gradient flow converges to an in-context OLS gradient-descent step (orthogonal scope: toy linear, no MoE/deployment-gap).

MoE backdoors and defences. BadMoE (Anonymous, 2025a) reports dormant-expert poisoning at low rates with optimised routing triggers; BadPatches (Anonymous, 2025b) extends to vision MoE patch triggers; BadSwitch (Liu et al., 2025) (NeurIPS 2025) uses task-coupled trigger optimization plus Top-S expert tracing; GateBreaker (Anonymous, 2026b) (USENIX Security 2026) uses training-free gate profiling for inference-time attacks. Defences SafeMoE (Anonymous, 2025d) (penalises routing drift to preserve safety) and SAFEx (Anonymous, 2025e) (NeurIPS 2025, stability-based safety-expert identification) empirically confirm the routing-drift mechanism but stop short of a closed-form threshold. Routing-aware safety alignment frameworks like RASA (Anonymous, 2026c) are a repair target, not a diagnostic. Re-MoE (Anonymous, 2025c) replaces the kink with differentiable routing, removing the bifurcation regime entirely; we deliberately keep hard top- K to make the structural mechanism explicit. Routing-structure analyses for domain specialisation (Anonymous, 2026a) address committee-vs-specialised routing, which is disjoint from the safety-trigger response we study.

Influence functions and matched-estimand response. The $H^{-1}g$ object in Theorems 4.3–4.4 is the linear-response operator at the heart of influence functions (Koh & Liang, 2017). The empirical gap between IF prediction and actual leave-one-out retraining is formalised by Bae et al. (2022): IFs predict the Proximal Bregman Response Function (PBR), distinct from finite-step retraining response by warm-start, proximity, and non-convergence terms; Schioppa et al. (2023) extend this to finite training trajectories. Park et al. (2023) (TRAK) and Kwon et al. (2024) (DataInf) provide scalable IF/data-attribution methodology for LoRA-tuned LLMs and linearised models. PALR is the AdamW matched estimand in this lineage: the finite- K optimizer susceptibility that population IFT predicts only in the converged-vanilla-GD limit (App. C).

Singular learning theory. Watanabe SLT (Watanabe, 2009) provides the closest basin-crossover machinery, used here as auxiliary contextualisation rather than the primary basis. The line of empirical SLT-on-NN work (Hoogland et al., 2024) has not yet been extended to MoE routing.